

QA STA642 100% VERIFIED
ASSESSMENT (108 - 187) + CURRENT
QUIZ# 3 + OBJECTIVE + SUBJECTIVE
SOLVED BY IBR

A handwritten signature in black ink, appearing to be 'ing' followed by a stylized flourish.

STA642 VERIFIED ASSESSMENT (108 - 187)

Time Left: 84 sec(s)

Question # 1 of 1

A representative from an ice-cream producing company randomly stops people on a randomly selected street in the city of Lahore until he finds a person who actually purchased and tried the latest product of this company meaning a new kind of ice-cream they have just now brought into the market. The probability of success $p = 0.10$ then what is the mean and variance of the Geometric distribution?

Select the correct option

- ☐ $\mu = 90 \text{ and } \sigma^2 = 10$
- ☐ $\mu = 1 \text{ and } \sigma^2 = 9$
- ☒ $\mu = 10 \text{ and } \sigma^2 = 90$
- ☐ $\mu = 10 \text{ and } \sigma^2 = 10$

we are given that
 $p = 0.10$
we have to find
mean and variance of Geometric Dist

as we know that
mean $= \mu = \frac{1}{p} = \frac{1}{0.10} = 10$
variance $= \sigma^2 = \frac{1-p}{p^2}$
 $= \frac{1-0.10}{(0.10)^2} = \frac{0.9}{0.01} = 90$

$\mu = 10$, $\sigma^2 = 90$

 Reload Math Equations

Question # 1 of 1

Time Left: 85 sec(s)

In hypergeometric distribution, the successive trials are

Select the correct option

 Reload Math Equations

☐	independent	
☒	dependent	✓
☐	not fixed	
☐	fixed	

The parameters of the hypergeometric distribution are

Select the correct option

 Reload Math Equations

☐	4	
☒	3	✓ n, k, N
☐	2	
☐	1	

Question # 1 of 1

Time Left: 84 sec(s)

The mean of hypergeometric distribution is

Select the correct option

 Reload Math Equations

☐

np

☐

nk

☒

nk/N

☐

np/N

Question # 1 of 2

Time Left: 51 sec(s)

The parameter of Poisson distribution is

Select the correct option

 Reload Math Equations

☐	none of the above	
☐	σ	
☒	μ	✓
☐	both	

Question # 2 of 2

Time Left: 64 sec(s)

The recurrence relation between $P(x)$ and $P(x+1)$ in a Poisson distribution is given by _____

Select the correct option

 Reload Math Equations

$$m P(x+1) - P(x) = 0$$



$$(x+1) P(x) - x P(x+1) = 0$$



$$P(x+1) - m P(x) = 0$$



$$(x+1) P(x+1) - m P(x) = 0$$



Question # 1 of 1

Time Left: 83 sec(s)

The p.d.f of Poisson Distribution is given by _____

Select the correct option

 Reload Math Equations

☒	$\frac{e^{-m} m^x}{x!}$	✓
☐	$\frac{e^{-m} x!}{m^x}$	
☐	$\frac{x!}{m^x e^{-m}}$	
☐	$\frac{e^m m^x}{x!}$	

Question # 1 of 1**Time Left: 71 sec(s)**

In a Poisson Distribution, if 'n' is the number of trials and 'p' is the probability of success, then the mean value is given by?

Select the correct option[Reload Math Equations](#)

☐	$m=np(1-p)$	
☐	$m=(np)^2$	
☒	$m=p$	
☐	$m=np$	✓

Question # 1 of 2

Time Left: 82 sec(s)

For a Poisson Distribution, if mean(m) = 1, then $P(1)$ is?

Select the correct option

☐ $1/e$ ☐

Indeterminate

☐ e ☐ $e/2$

i.f m
mean(m) = 1, then $P(1) = ?$
as we know that

$$P(x=x) = \frac{e^{-m} m^x}{x!}$$
$$P(1) = \frac{e^{-1} (1)^1}{1!} = e^{-1} = \frac{1}{e}$$

Reload Math Equations

Question # 2 of 2

Time Left: 86 sec(s)

If 'm' is the mean of a Poisson Distribution, then variance is given by _____

Select the correct option

 Reload Math Equations

☐	m^2	✓ mean=variance m=m
☐	$m^{1/2}$	
☐	$m/2$	
☒	m	

Question # 1 of 1

Time Left: 80 sec(s)

A student tries to break his pencil of unit length two pieces uniformly. The expected length of the shorter piece of pencil is.....

Select the correct option

- ☐ 1/8
- ☐ 1/16
- ☐ 1/2
- ☐ 1/4



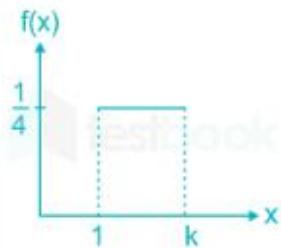
Here x is uniformly distributed
between 0 to $\frac{1}{2}$
i.e. $x \in [0, \frac{1}{2}]$

Then pdf will be
 $f(x) = \frac{1}{(\frac{1}{2} - 0)} = \frac{1}{\frac{1}{2}} = 2$

Expected length $E(X)$ is
 $= \int_0^{\frac{1}{2}} x f(x) dx$
 $= \int_0^{\frac{1}{2}} x \cdot 2 dx = 2 \int_0^{\frac{1}{2}} x dx$
 $= 2 \left[\frac{x^2}{2} \right]_0^{\frac{1}{2}} = 2 \left[\frac{(\frac{1}{2})^2}{2} - 0 \right]$
 $= 2 \left[\frac{\frac{1}{4}}{2} \right] = 2 \left[\frac{1}{8} \right] = \frac{1}{4}$

[Reload Math Equations](#)

The continuous random variable X has pdf $f(x)$ as shown below. Find the variance?



Select the correct option

[Reload Math Equations](#)

☒	$\frac{4}{3}$	✓
☐	$\frac{2}{\sqrt{3}}$	
☐	$\frac{16}{9}$	
☐	$\frac{5}{3}$	

Question # 1 of 1

Time Left: 86 sec(s)

The shape parameters of the Gamma distribution are

Select the correct option

 Reload Math Equations

☐

β

☐

α

☐

γ

☐

λ

Question # 1 of 1

Time Left: 85 sec(s)

The parameter of Gamma distribution is /are:

Select the correct option

 Reload Math Equations

- | | | |
|----------------------------------|---|---|
| ☐ | 3 | |
| ☒ | 2 | ✓ |
| ☐ | 4 | |
| ☐ | 1 | |

Question # 1 of 1

Time Left: 85 sec(s)

The scale parameters of gamma distribution are

Select the correct option

 Reload Math Equations

☐

γ

☐

θ

☒

β

☐

α

Question # 1 of 1

Time Left: 83 sec(s)

The mean of the gamma distribution is

Select the correct option

 Reload Math Equations

☐	$\alpha\gamma$	
☐	$\alpha\beta^2$	
☐	$\alpha^2\beta$	
☒	$\alpha\beta$	✓

Question # 1 of 1

Time Left: 81 sec(s)

The shape of normal distribution is

Select the correct option

 Reload Math Equations

☐

b and c both

☐

asymmetrical

☒

symmetrical

☐

bell shaped

Question # 1 of 1

Time Left: 57 sec(s)

Total area under the density curve

Select the correct option

 Reload Math Equations

☒	1	✓
☐	0.5	
☐	0.25	
☐	0	

Question # 1 of 1

Time Left: 84 sec(s)

Normal Distribution is applied for _____

Select the correct option

 Reload Math Equations

☒	Discrete Random Variable	
☐	Irregular Random Variable	
☐	Uncertain Random Variable	
☐	Continuous Random Distribution	✓

Question # 1 of 1

Time Left: 82 sec(s)

Normal Distribution is also known as _____

Select the correct option

 Reload Math Equations

☐	Laplacian Distribution	
☐	Cauchy's Distribution	
☒	Lagrangian Distribution	
☐	Gaussian Distribution	✓

Question # 1 of 1

Time Left: 85 sec(s)

The standard normal curve is symmetric about the value _____

Select the correct option

 Reload Math Equations

☐	1	
☒	0	✓
☐	None	
☐	0.5	

Question # 1 of 1

Time Left: 86 sec(s)

For a normal distribution its mean, median, mode are equal.

Select the correct option

 Reload Math Equations

True

☒

False

☐

Question # 1 of 1

Time Left: 84 sec(s)

For a standard normal variate, the value of Standard Deviation is _____

Select the correct option

 Reload Math Equations

☐	0	
☐	infinity	
☐	1	✓
☐	not defined	

Question # 1 of 1

Time Left: 77 sec(s)

In Normal distribution, the highest value of ordinate occurs at _____

Select the correct option

 Reload Math Equations

☒	Mean	✓
☐	Extremes	
☐	Variance	
☐	Same value occurs at all points	

Normal Distribution is symmetric is about _____

Select the correct option

 Reload Math Equations

- | | | |
|-----------------------|--------------------|---|
| ☐ | Variance | |
| ☐ | Covariance | |
| ☐ | Standard deviation | |
| ☐ | Mean | ✓ |

Question # 1 of 1

Time Left: 85 sec(s)

For a standard normal variate, the value of mean is?

Select the correct option

 Reload Math Equations

☒	0 ✓
☐	None
☐	1
☐	∞

Question # 1 of 1

Time Left: 61 sec(s)

The relationship between mean deviation and standard deviation is

Select the correct option

 Reload Math Equations

☐	Nonw	
☐	$M.D = 3/4(S.D)$	
☒	$M.D = 4/5(S.D)$	✓
☐	$M.D = 1/2(S.D)$	

Question # 1 of 2

Time Left: 88 sec(s)

All odd order moments in normal distribution about the mean are

Select the correct option

 Reload Math Equations

☐	Undefined	
☐	zero	✓
☐	infinity	
☐	None of the above	

QA STA642 VERIFIED QUIZ#3

STA642 - Probability Distributions (Quiz 3)

sec(s)

Quiz Start Time: 12:21 PM, 26 August 2022

Question # 1 of 5 (Start time: 12:21:42 PM, 26 August 2022)

Total Marks: 1

Poisson distribution is applied for _____

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|----------------------------|
| ☒ | Discrete Random Variable |
| ☐ | Continuous Random Variable |
| ☐ | Irregular Random Variable |
| ☐ | Uncertain Random Variable |

Click to Save Answer & Move to Next Question

Question # 1 of 5 (start time: 01:47:50 PM, 26 August 2022)

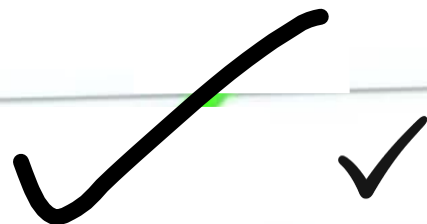
Total Marks: 1

Mean and variance of the poisson distribution are

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|--------------|
| ☐ | greater than |
| ☒ | equal |
| ☐ | less than |
| ☐ | not equal |

 Click to Save Answer & Move to Next Question

Question # 2 of 5 (Start time: 12:23:10 PM, 26 August 2022)

Total Marks: 1

Gamma function is defined as $\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$?

Select the correct option

 Reload Math Equations

☐	False
☒	True

[Click to Save Answer & Move to Next Question](#)

Question # 3 of 5 (start time: 12:24:10 PM, 26 August 2022)

Total Marks: 1

Which of the following graph represents gamma distribution?

select the correct option

[Reload Math Equations](#)

Figure b

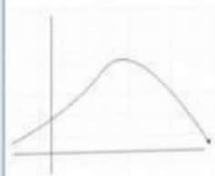


Figure d

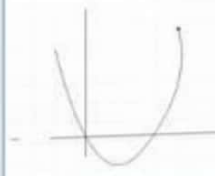
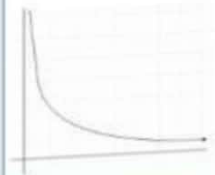


Figure c



Figure a

[Click to Save Answer & Move to Next Question](#)

Question # 5 of 5 (Start time: 12:27:00 PM, 26 August 2022)

Total Marks: 1

Exponential distribution is a Bivariate distribution.

Select the correct option



Reload Math Equations

☐	False
☒	True

Question # 3 of 5 (start time: 01:50:03 PM, 26 August 2022)

Total Marks: 1

The mean of exponential distribution is given as _____

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|-------------------|
| ☐ | $(\mu)^2$ |
| ☐ | $\frac{1}{\mu^2}$ |
| ☒ | μ |
| ☐ | $\frac{1}{\mu}$ |



Click to Save Answer & Move to Next Question

Question # 5 of 5 (start time: 03:46:21 PM, 26 August 2022)

Total Marks: 1

if

$$f_X(x) = \begin{cases} \frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

then on the value of $\alpha = 1$, which distribution we obtained?

Select the correct option

 Reload Math Equations

☐	Normal
☐	Exponential
☐	Poisson
☐	None



Question # 4 of 5 (Start time: 10:56:06 AM, 26 August 2022)

Total Marks: 1

The pdf of poisson distribution is

Select the correct option

Reload Math Equations

- | | |
|----------------------------------|-------------------------|
| ☐ | $\frac{e^{-m}m^x}{x!}$ |
| ☒ | $\frac{e^{-m}m^x}{x!}$ |
| ☐ | $\frac{e^{-m}x!}{m^x}$ |
| ☐ | $\frac{x!}{m^x e^{-m}}$ |



Click to Save Answer & Move to Next Question

Question # 5 of 5 (Start time: 10:58:30 AM, 26 August 2022)

Total Marks: 1

The mean and variance of the Gamma distribution are

Select the correct option

Reload Math Equations

- | | |
|----------------------------------|--|
| ☒ | $E(x) = \frac{\alpha}{\lambda} \text{ and } Var(x) = \frac{\alpha}{\lambda^2}$ |
| ☐ | $E(x) = \frac{1}{\lambda} \text{ and } Var(x) = \frac{\alpha}{\lambda^2}$ |
| ☐ | $E(x) = \frac{\alpha}{\lambda} \text{ and } Var(x) = \frac{1}{\lambda^2}$ |
| ☐ | $E(x) = \frac{\alpha+\beta}{\lambda} \text{ and } Var(x) = \frac{\alpha}{\lambda^2}$ |



Click to Save Answer & Move to Next Question

STA642 – Probability Distributions (Quiz 3)

Question # 5 of 5 (Start time: 10:06:06 PM, 26 August 2022)

The mean and variance of the Gamma distribution are

Select the correct option

☐	$E(x) = \frac{\alpha}{\lambda} \text{ and } Var(x) = \frac{\alpha}{\lambda^2}$	✓
☐	$E(x) = \frac{\alpha * \beta}{\lambda} \text{ and } Var(x) = \frac{\alpha}{\lambda^2}$	
☐	$E(x) = \frac{\alpha}{\lambda} \text{ and } Var(x) = \frac{1}{\lambda^2}$	
☐	$E(x) = \frac{1}{\lambda} \text{ and } Var(x) = \frac{\alpha}{\lambda^2}$	

Question # 1 of 5 (Start time: 09:11:58 PM, 26 August 2022)

Total Marks: 1

Sum of n independent Exponential random variables (λ) results in _____

Select the correct option



Reload Math Equations

☐	Uniform random variable
☒	Gamma random variable
☐	Binomial random variable
☐	Normal random variable



Click to Save Answer & Move to Next Question

Question # 1 of 5 (Start time: 03:43:15 PM, 26 August 2022)

Total Marks: 1

If we increase the value of $\alpha = 10^5$ with the $\lambda = 1/2$ then the shape of the distribution turns in to ----- distribution.

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---------------------|
| ☒ | Normal |
| ☐ | flatter than before |
| ☐ | positively skewed |
| ☐ | negatively skewed |

Click to Save Answer & Move to Next Question

Question # 2 of 5 (Start time: 08:42:22 AM, 27 August 2022)

Consider a random variable with exponential distribution with $\lambda=1$. Compute the probability for $P(X>3)$?

Select the correct option



- | | |
|-----------------------|----------|
| ☐ | e^{-4} |
| ☐ | e^{-1} |
| ☐ | e^{-3} |
| ☐ | e^{-2} |

✓ e^{-3}

Click to Save Answer & Move

①

we are given that

$$\lambda = 1$$

we have to find

$$P(X > 3)$$

as we know that $\Rightarrow f(x) = \lambda e^{-\lambda x}$

$$P(X > 3) = \int_3^{\infty} (1) e^{-x} dx$$

$$= \left[-e^{-x} \right]_3^{\infty} = \left[-e^{-\infty} - (-e^{-3}) \right]$$

$$= 0 + e^{-3} = e^{-3}$$

$$\boxed{P(X > 3) = e^{-3}}$$

Question # 4 of 5 (Start time: 12:25:42 PM, 26 August 2022)

Total Marks: 1

Job arrive at a facility at an average rate of 5 in an 8 hour shift. The arrival of the jobs follows poisson distribution. The average service time of the job on the facility is 40 minutes. The service time follows an exponential distribution. Idle time(in hours) at a facility per shift will be

Select the correct option

 Reload Math Equations

☐	$\frac{88}{3}$
☐	$\frac{88}{3}$
☒	$\frac{14}{3}$
☐	$\frac{14}{3}$



14/3

Click to Save Answer & Move to Next Question

Jobs arrival rate follows Poisson distribution

$$\text{Arrival rate } (\lambda) = \frac{5}{8} \left(\frac{\text{Jobs}}{\text{Hours}} \right)$$

Service time for one job is 40 min

$$\text{Therefore, Service rate } (\mu) = \frac{3}{2} \left(\frac{\text{Jobs}}{\text{Hour}} \right)$$

$$\text{Utilization factor} = \rho = \frac{\lambda}{\mu}$$

$$\text{Idle time} = 1 - \rho$$

now

$$\lambda = \frac{5}{8}, \mu = \frac{3}{2}, \rho = \frac{\lambda}{\mu} = \frac{\frac{5}{8}}{\frac{3}{2}} = \frac{5}{12}$$

$$\text{Idle time} = 1 - \rho = 1 - \frac{5}{12} = \frac{7}{12} \text{ hours}$$

Therefore, Idle time for 8-hours shift:

$$\frac{7}{12} \times 8^2 = \boxed{\frac{14}{3}}$$

STA642 - Probability Distributions (Quiz 3)

Question # 4 of 5 (Start time: 01:50:13 PM, 26 August 2022)

In a Poisson Distribution, If mean $(m) = e$, then $P(x)$ is given by -----

Select the correct option

☐	$\frac{e^{(m-x)}}{x!}$
☐	$\frac{x!}{e^{(x-m)}}$
☒	$\frac{e^{(x-m)}}{x!}$
☐	$\frac{x!}{e^{(m-x)}}$



(4)

we are given that
mean (m) = e

as we know that

$$P(x) = \frac{e^{-\mu} \mu^x}{x!}$$

where $\mu = \text{mean} = e = m$

so

$$P(x) = \frac{e^{-e} e^x}{x!} = \frac{e^{(-e+x)}}{x!}$$

$$P(x) = \frac{e^{(x-m)}}{x!}$$

$$\because e = m$$

Question # 2 of 5 (start time: 03:44:34 PM, 26 August 2022)

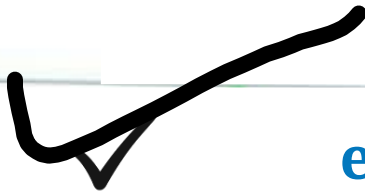
A random variable X has an exponential distribution with probability distribution function is given by $f(x) = 3e^{-3x}$ for $x > 0$

$= 0$ otherwise

Find probability that X is not less than 2?

Select the correct option

Relo

- | | |
|-----------------------|------------|
| ☐ | $e^{-6}-1$ |
| ☐ | e^{-6} |
| ☐ | $e^{-6}-3$ |
| ☐ | e^{-3} |
-  e^{-6}

Click to Save Answer & Move

2

$$f(x) = 3e^{-3x}$$

for $x > 0$
Find probability that x is not less than 2?

$$P(X > 2) = \int_2^{\infty} 3e^{-3x} dx$$

$$= -\frac{3}{3} \left[e^{-3x} \right]_2^{\infty}$$

$$= -1 \left[e^{-3(\infty)} - e^{-3(2)} \right]$$

$$= -1 (0 - e^{-6})$$

$$\boxed{P(X > 2) = e^{-6}}$$

Question # 5 of 5 (Start time: 09:33:43 PM, 26 August 2022)

If 'm' is the mean of Poisson Distribution, the $P(0)$ is given by _____

Select the correct option

☒	e^{-m}
☐	e
☐	e^m
☐	m^{-e}

given that .

mean = m

, $P(0) = ?$

we know that

$$P(x) = \frac{e^{-\mu} \mu^x}{x!}$$

$$P(0) = \frac{e^{-m} m^0}{0!} = \frac{e^{-m} (1)}{1}$$

$$\boxed{P(0) = e^{-m}}$$

Question # 4 of 5 (Start time: 05:21:09 PM, 26 August 2022)

Total Marks: 10

A mobile conversation follows an exponential distribution $f(x) = \frac{1}{3}e^{-\frac{x}{3}}$. What is the probability that the conversation takes more than 6 times?

Select the correct option

 Reload Math Equation

- | | |
|----------------------------------|-------------|
| ☐ | $e^{-3/3}$ |
| ☒ | $e^{-6/3}$ |
| ☐ | $e^{-1/3}$ |
| ☐ | $3e^{-1/3}$ |

Click to Solve Answer & Move to Next Question

$$f(x) = \frac{1}{3} e^{-x/3}$$

$$P(X > 5) = ?$$

$$P(X > 5) = \int_5^{\infty} \frac{1}{3} e^{-x/3} dx$$

$$= -\frac{3}{3} \left[e^{-x/3} \right]_5^{\infty}$$

$$= -1 \left[e^{-\infty/3} - e^{-5/3} \right]$$

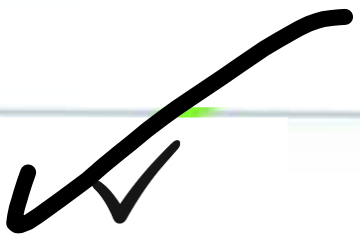
$$= -1 (0 - e^{-5/3})$$

$$\boxed{P(X > 5) = e^{-5/3}}$$

Question # 3 of 5 (Start time: 08:19:36 PM, 26 August 2022)

Find the value of $\Gamma(5/2)$?

Select the correct option

- 
- | | |
|-----------------------|--------------------------------|
| ☐ | $\frac{3}{4}\pi^{\frac{1}{2}}$ |
| ☐ | $\frac{7}{4}\pi^{\frac{1}{2}}$ |
| ☐ | $\frac{1}{4}\pi^{\frac{1}{2}}$ |
| ☐ | $\frac{5}{4}\pi^{\frac{1}{2}}$ |

Click to Save Answer & Move to Next Question

Find the value of $\Gamma(5/2) = ?$
as we know that ∞

$$\Gamma(\alpha) = \int_0^{\infty} y^{\alpha-1} e^{-y} dy \rightarrow (1)$$

This can be further simplified as follow:

$$\Gamma(\alpha) = (\alpha-1) \Gamma(\alpha-1) \rightarrow (2)$$

By using (2)

$$\begin{aligned}\Gamma(5/2) &= (\alpha-1) \Gamma(\alpha-1) \\ &= (5/2-1) \Gamma(5/2-1) \\ &= (3/2) \Gamma(3/2)\end{aligned}$$

Now by calculating

$$\Gamma(3/2)$$

$$\text{using } \Gamma(n+1) = n \Gamma(n)$$

$$\therefore \Gamma(1/2) = \sqrt{\pi}$$

$$\Gamma(n) = \frac{1}{n} \Gamma(n+1)$$

$$n = 3/2$$

$$\Gamma(3/2) = \Gamma(1/2 + 1)$$

$$= \frac{1}{2} \Gamma(1/2) = \frac{1}{2} \sqrt{\pi}$$

$$\Gamma(5/2) = \frac{3}{2} \Gamma(3/2) = \frac{3}{2} \left(\frac{1}{2} \sqrt{\pi} \right)$$

$$\Gamma(5/2) = \frac{3}{4} \sqrt{\pi} = \left(\frac{3}{4} (\pi)^{1/2} \right)$$

STA642 FINAL OBJECTIVE

#	Statement	Answer
1	Geometric distribution is a special case of which distribution	Negative binomial distribution
2	Instead of counting the number of successes we are interested to count the numbers of trails until a success is obtained. Which probability distribution should be used in this case	Binomial distribution
3	Choose the correct formula of mean of multinomial distribution	$E(X_i) = nP_i$
4	The domain $0 < x < y < 1$ can be Re-written as	$0 < x < 1, x < y < 1$
5	X and Y are independent if and only if	$P = 0$
6	In normal distribution (sigma square) denotes the	Variance
7	For the standard normal distribution E(z) is equal to which value	$E(z) = 0$
8	Choose the correct number of parameters(s) for normal distribution	μ, σ
9	What is relationship b/w the mean, median and mode of a normal distribution	Are equal
10	In normal distribution, the all odd order moments about mean are	Zero
11	The first Raw moment of the normal distribution is equal to	Expected value or mean
12	The first cumulant $k'(0)$ of normal distribution is	$K_1 = \mu$
13	All the higher cumulant of a normal distribution is	Zero
14	The bivariate normal distribution has	Two independent variable
15	The bivariate distribution can be expressed as	P_{iL}
16	Any subset of a sample space is called	Event
17	The quantile function is called	Percentile function or inverse of cumulative distribution
18	If the distribution satisfies the definition of “random variable” even though it does not Appear random in the everyday sense of the word; it is considered as	Degenerate
19	The expected value of e^{itx} is called	Characteristic function
20	Joint probability distribution of two variable is called	Bivariate distribution
22	In Which distribution the successive draws are conducted through sampling without replacement	Hyper geometric distribution
23	The mean and variance of the hyper geometric distribution is given	$\text{Mean} = E(X) = n \frac{K}{N}$ & $\text{Variance} = n \frac{K}{N} \frac{(N-K)}{N} \frac{N-n}{N-1}$
24	The mean and variance in poisson distribution is given by	mean and variance of the Poisson distribution are both equal to (lambda)

25	Which distribution has a unique property that its mean and variance are equal	Poisson
26	The formula $\frac{a+b}{2}$ is used to find the mean of which distribution	Continuous uniform distribution
27	For the exponential distribution, if $\theta=1/\lambda$ then θ ?	Give some value and find $\theta=1/\lambda$
28	The given pdf belongs to which distribution (Gamma distribution ka pdf tha)	Gamma distribution
29	For any fixed value of the scale parameter β as α increase, the shape of the distribution tends to	Normality shape
30	The limit of probability density function of the β -distribution are	$0 \leq x \leq 1$
31	The shape parameters of beta distribution is	α and β
32	The 2-parameters of uniform distribution are	a and b
33	Which two parameters determine the shape of binomial distribution	p and n
34	What is the formula of finding the variance of binomial distribution	$\sigma^2 = npq$
35	For the m^{th} moment about the origin if $m=1$ then we obtain	Mean
36	For the m^{th} moment about the origin if $m=2$ then we obtain	Variance
37	Consider the equation of circle $(x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$ which is showing the value of radius	a
38	How many parameters are in binomial normal distribution	two, n and p
39	A variable is normally distributed with $N(70, 100)$, what is the mean of this distribution	mean=70 $70 \Rightarrow \text{mean}, 100 \Rightarrow \text{variance}$
40	In the case of a Bivariate Normal distribution the random variables X and Y are independent If and only if the correlation coefficient ρ is equal to	zero

#	Statement	A	B	C	D
1	In one-to-one transformation, for two old variables, we get a	One new variable	Three new variable	Four new variable	Two new variable
2	The shape parameters of beta distribution is	$-\alpha$ and β	α and $-\beta$	$-\alpha$ and $-\beta$	α and β
3	Discrete probability distribution in which outcome is very small period is classified as a	Poisson distribution	Posterior distribution	Cumulative distribution	Normal distribution
4	The domain $0 < x < y < 1$ can be Re-written as	$0 < x < 1$, $x < y < 1$	$0 < x+y < 1$	$0 < y < 1$ $y < x < 1$	$0 < x < 1$ $y < x < 1$
6	The normal curve is asymptotic to which axis	x-axis	y-axis	z-axis	Both x-axis and y-axis
7	Which two parameter determine the shape of binomial distribution	p and n	p and x	p and q	q and x
9	In which distribution, the numbers of trial are not fixed in advance.	Binomial distribution	Bernoulli distribution	Hyper geometric distribution	Geometric distribution
11	For the standard normal distribution $E(z)$ is equal to which value,	1	0	-1	100
12	The limit of probability density function of the β -distribution are	$0 \leq x \leq 1$ $\alpha, \beta > 0$			
13	For the join PMF $P_{y_1 y_2}(y_1 y_2)$ we say obtain the marginal PMF of y_1 by summing on	Y_2			
14	Any subset of a sample space is called	Event			
15	Generally probability density function is symbolically denoted by	$F_x(x)$			
16	If random variable X is infinite then its expectation	Does not exist			
17	Chose the correct parameters of poisson distribution	λ			
18	The formula $\frac{a+b}{2}$ is used to find the mean of which	Continues uniform distribution			

	<i>distribution</i>	
19	The given PDF belongs to which distribution $f(x) = \begin{cases} \frac{1}{\alpha\beta^n} x^{n-1} e^{-\frac{x}{\beta}} & 0 < x < \infty, \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases}$	Gamma distribution
20	In which distribution the successive draws are conducted through sampling without replacement	Hyper geometric distribution
21	What is formula of finding the MEAN of binomial distribution	np
22	For any fixed value of the scale parameter β as α increase, the shape of the distribution tends to	Normality shape
23	X and Y are independent if and only if	F (x,y)=f (x)f (y)
24	Two parameter of uniform distribution are	a and b
25	What is the correct formula of mean of hyper geometric distribution	$\frac{nk}{N}$
26	The experiment continuous until r successive are observed where r is specified in advance. This is an example of which distribution	Negative binomial distribution
27	What is the relationship between mean, median, mode of a normal distribution	Always same
28	How many parameter are in the Bivariate normal distribution	Five
29	The biraviate normal distribution can be expressed as	$N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$
30	The bivariate normal distribution has	Bell shape
31	In normal distribution, the all odd order moment of about mean are	Zero
32	In general, the expected value of a product is _____ to the product of the expected value that is E(XY)____ E(X)E(Y)	Not equal , ≠
33	Which distribution pass the quality that is means and standard deviation are equal	Exponential distribution
34	The row moment of the normal distribution is equal to the	Mean E(X)
35	All the higher commulant of a normal distribution are equal to	Zero
36	In Bivariate Normal Distribution, what is the range limit of variable X and Y	$-\infty < x, y < \infty$
37	The first commulant $K'(0)$ of normal distribution is	Mean (μ)
38	In the case of Bivariate normal distribution, the random variable X and Y are independent iff the correlation coefficient (ρ) =	$\rho = 0$
39	Choose the correct formula of VARIENCE of multinomial distribution	$np_i (1 - p_i)$
40	Consider the equation of circle $(x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$	a

BINOMIAL DISTRIBUTION

MCQ 8.1

A Bernoulli trial has:

- (a) At least two outcomes (b) At most two outcomes
(c) **Two outcomes** (d) Fewer than two outcomes

MCQ 8.2

The two mutually exclusive outcomes in a Bernoulli trial are usually called:

- (a) **Success and failure** (b) Variable and constant
(c) Mean and variance (d) With and without replacement

MCQ 8.3

Nature of the binomial random variable X is:

- (a) Quantitative (b) Qualitative (c) **Discrete** (d) Continuous

MCQ 8.4

In a binomial probability distribution, the sum of probability of failure and probability of success is always:

- (a) Zero (b) Less than 0.5 (c) Greater than 0.5 (d) **One**

MCQ 8.5

In a binomial experiment, the successive trials are:

- (a) Dependent (b) **Independent** (c) Mutually exclusive (d) Fixed

MCQ 8.6

The parameters of the binomial distribution are:

- (a) **n and p** (b) p and q (c) np and nq (d) np and npq

MCQ 8.7

The range of binomial distribution is:

- (a) **0 to n** (b) 0 to ∞ (c) -1 to +1 (d) 0 to 1

MCQ 8.8

The mean and standard deviation of the binomial probability distribution are respectively:

- (a) np and npq (b) **np and $\sqrt{npq}$** (c) np and nq (d) n and p

MCQ 8.9

In a binomial experiment with three trials, the variable can take:

- (a) 2 values (b) 3 values (c) **4 values** (d) 5 values

not sure

MCQ 8.10

The shape of the binomial probability distribution depends upon the values of its:

- (a) Mean (b) Variance (c) **Parameters** (d) Quartiles

MCQ 8.11

In binomial distribution the numbers of trials are:

- (a) Very large (b) Very small (c) **Fixed** (d) Not fixed

MCQ 8.12

In a binomial probability distribution, relation between mean and variance is:

- (a) Mean < Variance (b) Mean = Variance
(c) **Mean > Variance** (d) Difficult to tell

MCQ 8.13

In binomial distribution when $n = 1$, then it becomes:

- (a) Hypergeometric distribution (b) Normal distribution
(c) Uniform distribution **(d) Bernoulli distribution**

MCQ 8.14

The mean of a binomial distribution depends on:

- (a) Number of trials (b) Probability of success
(c) Probability of failure **(d) Number of trials and probability of success**

MCQ 8.15

The variance of a binomial distribution depends on:

- (a) Number of trials (b) Probability of success
(c) Probability of failure **(d) All of the above**

MCQ 8.16

Which of the following is not property of a binomial experiment?

- (a) Probability of success remains constant
(b) n is fixed
(c) Successive trials are dependent
(d) It has two parameters

MCQ 8.17

The binomial probability distribution is symmetrical when:

- (a) $p = q$** (b) $p < q$ (c) $p > q$ (d) $np > npq$

MCQ 8.18

The binomial distribution is negatively skewed if:

- (a) $p < 1/2$ (b) $p = 1/2$ **(c) $p > 1/2$** (d) $p = 1$

MCQ 8.19

In a binomial probability distribution, the skewness is positive for:

- (a) $p < 1/2$** (b) $p = 1/4$ (c) $np = npq$ (d) $np = nq$

MCQ 8.20

Which of the following statements is false?

- (a) Expected value of a constant is constant
(b) In a binomial distribution the standard deviation is always less than its variance
(c) In a binomial distribution the mean is always greater than its variance
(d) In binomial experiment the probability of success remains constant from trial to trial

MCQ 8.21

If a binomial probability distribution has parameters $(n, p) = (5, 0.6)$, the probability of $x = 3.5$ is:

- (a) 0** (b) 1 (c) 0.6 (d) 0.4 impossible

:

MCQ 8.23

If $n = 6$ and $p = 0.9$ then the value of $P(x=7)$ is:

- (a) Zero** (b) Less than zero (c) More than zero (d) One

MCQ 8.24

In a binomial probability distribution, coefficient of skewness = $(q - p)/\sqrt{npq} = 0$, it means that the distribution is:

- (a) Symmetrical** (b) Skewed to the left (c) Skewed to the right (d) Highly skewed

MCQ 8.25

For a binomial distribution with $n = 10$, $p = 0.5$, the probability of zero or more successes is:

- (a) 1** (b) 0.5 (c) 0.25 (d) 0.75

MCQ 8.26

In a binomial distribution, the mean, median and mode coincide when:

- (a) $p < 1/2$ (b) $p > 1/2$ (c) $p \neq 1/2$ **(d) $p = 1/2$**

MCQ 8.27

In which distribution, the probability success remains constant from trial to trial?

- (a) Hypergeometric distribution **(b) Binomial distribution**
(c) Sampling distribution (d) Frequency distribution

MCQ 8.28

In a binomial experiment when $n = 5$, the maximum number of successes will be:

- (a) 0 (b) 2.5 (c) 4 **(d) 5**

MCQ 8.29

In a binomial experiment when $n = 10$, the minimum number of successes will be:

- (a) 0** (b) 5 (c) 10 (d) 11

MCQ 8.30

If $n = 10$ and $p = 0.6$, then $P(x \geq 0)$ is:

- (a) 0.5 (b) 0.6 **(c) 1.0** (d) 1.2

MCQ 8.31

A random variable X has a binomial distribution with $n = 4$, the standard deviation of X is:

- (a) $4pq$ **(b) $2\sqrt{pq}$** (c) $4p$ (d) $4(q+p)$

MCQ 8.32

In a multiple choice test there are five possible answers to each of 20 questions. If a candidate guesses the correct answer each time, the mean number of correct answers is:

- (a) 4** (b) 5 (c) $1/5$ (d) 20

MCQ 8.33

If three coins are tossed, the probability of two heads is:

- (a) $1/8$ **(b) $3/8$** (c) $2/3$ (d) 0

MCQ 8.35

The value of second moment about the mean in a binomial distribution is 36. The value of the standard deviation of a binomial distribution is:

- (a) 36 **(b) 6** (c) $1/36$ (d) $1/6$

MCQ 8.36

For a binomial probability distribution, the expected frequency of x successes in N experiments is:

- (a) $\frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}$ (b) $p^x(1-p)^{1-x}$
 (c) $N \binom{n}{x} p^x q^{n-x}$ (d) $\binom{n}{x} p^x(1-p)^{n-x}$

MCQ 8.37

In a binomial frequency distribution $100 (1/5 + 4/5)^5$. The parameters n and p are respectively:

- (a) (5, 1/5) (b) (1/5, 4/5) (c) (5, 4/5) (d) (100, 4/5)

MCQ 8.38

For a binomial frequency distribution $100 (1/5 + 4/5)$, the mean is:

- (a) 1/5 (b) 4/5 (c) 5 (d) 4

as know that
 $N(q+p)^n$
 so
 $mean = np$
 $mean = (5)(4/5)$
 $mean = 4$

HYPERGEOMETRIC DISTRIBUTION

MCQ 8.40

The hypergeometric distribution has:

- (a) One parameter (b) Two parameters (c) Three parameters (d) Four parameters

MCQ 8.41

The parameters of the hypergeometric distribution are:

- (a) N, n, p (b) N, n, np (c) N, n, k (d) n and p

MCQ 8.42

Nature of the Hypergeometric random variable is:

- (a) Continuous (b) Discrete (c) Qualitative (d) Quantitative

MCQ 8.43

In hypergeometric distribution, the successive trials are:

- (a) Independent (b) Dependent (c) Very large (d) Very small

MCQ 8.44

In a hypergeometric distribution, the probability of success:

- (a) Remains constant from trial to trial
 (b) Does not remain constant from trial to trial
 (c) Equal to probability of failure
 (d) Less than probability of failure

MCQ 8.45

If in a hypergeometric distribution $N = 10, k = 5$ and $n = 4$; then the probability of failure is:

- (a) 2 (b) 5 (c) 1 (d) 0.25 $(= N - k = 10 - 5 = 5)$

MCQ 8.46

The rang of hypergeometric distribution is:

- (a) 0 to n (b) 0 to k (c) 0 to N (d) 0 to n or k (whichever is less)

MCQ 8.47

The number of trials in hypergeometric distribution is:

- (a) Not fixed (b) Fixed (c) Large (d) Small

MCQ 8.48

The probability of a success changes from trial to trial in:

- (a) Binomial distribution
 (b) **Hypergeometric distribution**
 (c) Normal distribution
 (d) Frequency distribution

MCQ 8.49

The mean of the hypergeometric distribution is:

- (a) **$\frac{nk}{N}$**
 (b) $\frac{Nk}{n}$
 (c) $\frac{Nn}{k}$
 (d) $\frac{n+k}{N}$

MCQ 8.50

The standard deviation of the hypergeometric distribution is:

- (a) $\frac{nk(k-N)(N-n)}{N^2(N-1)}$
 (b) **$\sqrt{\frac{nk(N-k)(N-n)}{N^2(N-1)}}$**
 (c) $\sqrt{\frac{nk(N-k)(n-N)}{N^2(N-1)}}$
 (d) $\sqrt{\frac{nk(N-k) + (N-n)}{N^2(N-1)}}$

MCQ 8.51

In hypergeometric probability distribution, the relation between mean and variance is:

- (a) **Mean > variance**
 (b) Mean < Variance
 (c) Mean = Variance
 (d) Mean = 2Variance

MCQ 8.52

Which of the following is the property of hypergeometric experiment?

- (a) **p** remains constant from trial to trial
 (b) Successive trials are independent
 (c) **Sampling is performed without replacement**
 (d) **n** is not fixed

MCQ 8.53

Hypergeometric distribution reduces to binomial distribution when:

- (a) $N = n$
 (b) $n \rightarrow \infty$
 (c) **$N \rightarrow \infty$**
 (d) $N < n$

MCQ 8.54

In a hypergeometric distribution $N=6$, $n=4$ and $k=3$, then the mean is equal to:

- (a) **2**
 (b) 4
 (c) 6
 (d) 24
- $(\text{mean} = \frac{nk}{N} = \frac{4 \cdot 3}{6} = \frac{12}{6} = 2)$

MCQ 8.55

Given $N = 11$, $n = 5$, $k = 7$; $P(x \geq 1)$ equals:

- (a) **1**
 (b) $1/66$
 (c) $65/66$
 (d) None of the above

MCQ 8.56

Given $N=11$, $n=5$, $k=4$; $P(x \leq 4)$ equals:

- (a) Less than one
 (b) **Exactly one**
 (c) More than one
 (d) Between 0.5 and 1

$$k = 4$$

$$N - k = 11 - 4 = 7 \text{ (black beads)}$$

$$x = 0, 1, 2, 3, 4$$

$$P(X = x) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}$$

$$P(X = 0) = \frac{\binom{4}{0} \binom{7}{5}}{\binom{11}{5}} = \frac{(1)(21)}{462} = \frac{21}{462}$$

$$P(X = 1) = \frac{\binom{4}{1} \binom{7}{4}}{\binom{11}{5}} = \frac{(4)(35)}{462} = \frac{140}{462}$$

$$P(X = 2) = \frac{\binom{4}{2} \binom{7}{3}}{\binom{11}{5}} = \frac{(6)(35)}{462} = \frac{210}{462}$$

$$P(X = 3) = \frac{\binom{4}{3} \binom{7}{2}}{\binom{11}{5}} = \frac{(4)(21)}{462} = \frac{84}{462}$$

$$P(X = 4) = \frac{\binom{4}{4} \binom{7}{1}}{\binom{11}{5}} = \frac{(1)(7)}{462} = \frac{7}{462}$$

x	$f(x)$	
0	21/462	
1	140/462	
2	210/462	
3	84/462	
4	7/462	
Total	462/462=1	

QA STA642 FINAL TERM SUBJECTIVE SOLVED BY IBR

DISCRETE DISTRIBUTION

Arrangement of all possible values of random variable 'X' $x_1, x_2, \dots, x_n$ along with their respective probabilities $p(x_1), p(x_2), \dots, p(x_n)$ is called discrete probability distribution, having properties:

- ❖ $p(x) \geq 0$
- ❖ $0 \leq p(x) \leq 1$
- ❖ $\sum_{i=1}^n p(x_i) = 1$

Important discrete probability distribution are

1. Binomial distribution
2. Hypergeometric distribution
3. Poison distribution
4. Multinomial distribution
5. Negative binomial distribution
6. Geometric distribution
7. Gamma distribution
8. Beta distribution
9. Normal distribution
10. Exponential distribution
11. Uniform distribution

BINOMIAL DISTRIBUTION

Binomial Experiment

An experiment is called a **binomial experiment** if it possesses the following **four** properties:

1. The outcome of each trial may be classified into one of two categories, conventionally called **success (S)** and **Failure (F)**.

It is to be noted that the outcome of **interest** is called a **success** and the other, a **failure**.

2. The probability of success, denoted by p ; remains **constant** for all trials.
3. The successive trials are all **independent**.
4. The experiment is repeated a **fixed** number of times, say n .

Binomial Distribution

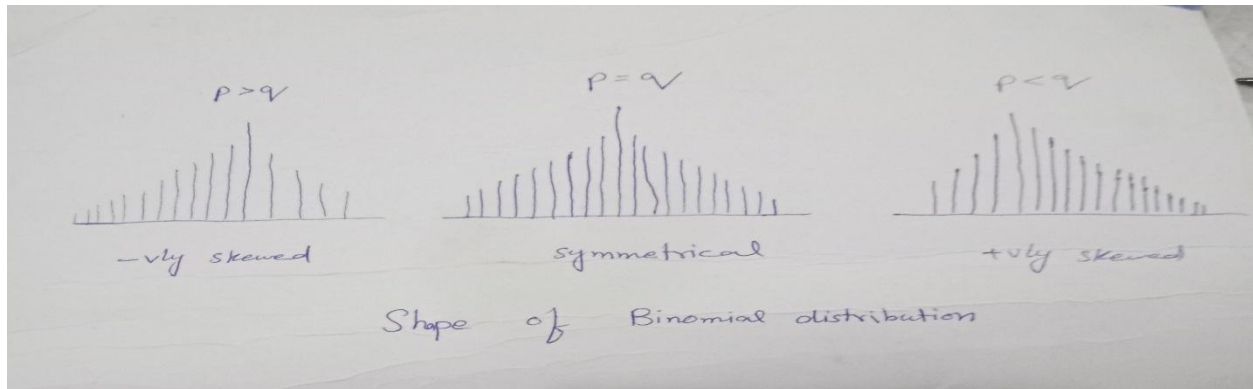
Let the random variable X denote the number of successes in n trials --- when a binomial experiment is performed.

Then the PMF of X is given by

$$p(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & x = 0, 1, 2, \dots, n \\ 0 & \text{elsewhere.} \end{cases} \quad (1)$$

where p is the probability of success in each trial and n is the number of trials.

SHAPE OF BINOMIAL DISTRIBUTION



- The **shape** of the binomial probability distribution depends on the values of the **two parameters** p and n .

Properties of binomial distribution

1. Binomial distribution is discrete by nature.
2. Binomial distribution having range of success $0, 1, 2, \dots, n$
3. It has two parameter shape of distribution depends n, p .
4. Area under binomial probabilities is unit i.e. $\sum p(x_i) = 1$.
5. Mean and variance of binomial are np and npq respectively.
6. In binomial distribution $np > npq$ (mean > variance).
7. As $n \rightarrow \infty$ binomial tends to normal distribution and poisson.

8. Its probability generating function is

$$G(\theta) = (p\theta + q)^n$$

9. Its moment generating function is

$$M_o(\theta) = (pe^t + q)^n$$

10. Its characteristic function of e^{itx}

$$\theta(t) = (pe^{it} + q)^n$$

11. If

p=q=0.5 distribution is symmetrical

p>q distribution is -ve skewed

p<q distribution is +ve skewed

12. Its coefficient of skewness and coefficient of kurtosis are

$$\beta_1 = \frac{(q-p)^2}{npq} \quad ; \quad \beta_2 = 3 + \frac{1-6pq}{npq}$$

PROBLEMS OF BINOMIAL DISTRIBUTION

Date: _____

BINOMIAL DISTRIBUTION

Q No 1:

An event has the probability $p = \frac{3}{8}$
Find the complete binomial distribution
for $n = 5$ trials.

Solution

We are given that

$$p = \frac{3}{8}, \quad n = 5, \quad x = 0, 1, 2, 3, 4, 5$$

$$q = 1 - p = 1 - \frac{3}{8} = \frac{5}{8}$$

as we know that

The pdf of binomial distribution is

$$P(X=x) = \binom{n}{x} p^x q^{n-x} \quad \text{for } x = 0, 1, 2, \dots, n$$

So

x

$P(X=x)$

$$0 \quad \binom{5}{0} \left(\frac{3}{8}\right)^0 \left(\frac{5}{8}\right)^{5-0} = 3125/32768$$

$$1 \quad \binom{5}{1} \left(\frac{3}{8}\right)^1 \left(\frac{5}{8}\right)^{5-1} = 9375/32768$$

$$2 \quad \binom{5}{2} \left(\frac{3}{8}\right)^2 \left(\frac{5}{8}\right)^{5-2} = 11250/32768$$

$$3 \quad \binom{5}{3} \left(\frac{3}{8}\right)^3 \left(\frac{5}{8}\right)^{5-3} = 6750/32768$$

$$4 \quad \binom{5}{4} \left(\frac{3}{8}\right)^4 \left(\frac{5}{8}\right)^{5-4} = 2025/32768$$

$$5 \quad \binom{5}{5} \left(\frac{3}{8}\right)^5 \left(\frac{5}{8}\right)^{5-5} = 243/32768$$

$$\frac{3125 + 9375 + 11250 + 6750 + 2025 + 243}{32768} = 1$$

Date: _____

QNo2.

If Y is $b(n, 1/3)$, then $P(Y \geq 1) = 1 - P(Y=0) = 1 - (2/3)^n$.

Find the smallest value of n that yields $P(Y \geq 1) > 0.80$

Solution

We are given that

$$b(n, 1/3), \text{ here } n=n, p=1/3$$

$$P(Y \geq 1) = 1 - P(Y=0) = 1 - (2/3)^n$$

We have to find $n=?$

$$P(Y \geq 1) > 0.80$$

$$1 - (2/3)^n > 0.80$$

$$1 - 0.80 > (2/3)^n$$

$$0.2 > (2/3)^n$$

By taking natural log on both sides

$$\ln(0.2) > n \ln(2/3)$$

$$-1.60943 > n(-0.4054)$$

$$n > 1.60943 / 0.4054$$

$$n > 3.97$$

Thus, we see that $\boxed{n=4}$ is the solution

Date: _____

Q203

The probability that a planted radish seed germinates is 0.80. A gardener plants nine seeds.
i) what is the average number of seeds?
ii) what is the standard deviation of seeds?

Solution:

we are given that

$$n = 9, p = 0.80, q = 1 - 0.80 = 0.20$$

as we know that

$$\begin{aligned}\text{average} = \text{mean} &= np \\ &= (9)(0.80) \\ &= 7.2\end{aligned}$$

$$\begin{aligned}\text{Standard deviation} &= \sqrt{npq} \\ &= \sqrt{(9)(0.80)(0.20)} \\ &= 1.2\end{aligned}$$

Date: _____

Q204

Q7

$$m(t) = \left(\frac{2}{3} + \frac{1}{3}e^t\right)^5$$

then find mean and variance of binomial distribution.

Solution

Here

$$q = \frac{2}{3}, p = \frac{1}{3}, n = 5$$

then

$$\text{mean} = np = (5)\left(\frac{1}{3}\right) = \frac{5}{3}$$

$$\text{Variance} = npq = (5)\left(\frac{1}{3}\right)\left(\frac{2}{3}\right) = \frac{10}{9}$$

MGF of Binomial Distribution

As we know that

The pdf of binomial distribution is

$$P(X=x) = \binom{n}{x} p^x q^{n-x} \quad \text{for } x = 0, 1, 2, 3, \dots, n$$

Where, $q = 1 - p$

The MGF of the binomial distribution is denoted by $M(t)$ and given as

$$M(t) = E(e^{tx}) = \sum_{x=0}^n e^{tx} P(X=x)$$

$$M(t) = \sum_{x=0}^n e^{tx} \binom{n}{x} p^x q^{n-x}$$

$$M(t) = \sum_{x=0}^n \binom{n}{x} p^x e^{tx} q^{n-x}$$

$$M(t) = [pe^t + q]^n$$

OR

$$M(t) = [pe^t + (1-p)]^n$$

for all real values of t.

Mean

By differentiating the MGF with respect to t

$$\frac{d[M(t)]}{dt} = M'(t) = \frac{d}{dt}[pe^t + q]^n$$

$$M'(t) = n[pe^t + q]^{n-1} pe^t \rightarrow (1)$$

now

$$M'(0) = n[pe^0 + q]^{n-1} pe^0$$

$$M'(0) = n[p + q]^{n-1} p$$

$$M'(0) = n[1]^{n-1} p$$

where

$$p + q = 1$$

$$M'(0) = np$$

as we know that

$$\mu = M'(0)$$

so

$$\mu = np$$

Mean = np

Variance

By differentiating (1) with respect to t

$$M''(t) = \frac{d}{dt} n[p e^t + q]^{n-1} p e^t$$

$$M''(t) = n(n-1)[p e^t + q]^{n-2} (p e^t)^2 + n[p e^t + q]^{n-1} p e^t$$

now

$$M''(0) = n(n-1)[p e^0 + q]^{n-2} (p e^0)^2 + n[p e^0 + q]^{n-1} p e^0$$

$$M''(0) = n(n-1)[p + q]^{n-2} p^2 + n[p + q]^{n-1} p$$

$$M''(0) = n(n-1)[1]^{n-2} p^2 + n[1]^{n-1} p$$

where

$$p + q = 1$$

$$M''(0) = n(n-1) p^2 + n p$$

as we know that

$$\sigma^2 = M''(0) - \mu^2$$

so

$$\sigma^2 = n(n-1) p^2 + n p - (np)^2$$

$$\sigma^2 = n^2 p^2 - n p^2 + n p - n^2 p^2$$

$$\sigma^2 = n p - n p^2$$

$$\sigma^2 = n p (1 - p)$$

as we know that

$$p + q = 1$$

$$q = 1 - p$$

so

$$\sigma^2 = n p q$$

Variance = npq

HYPERGEOMETRIC DISTRIBUTION

The following conditions characterize the hypergeometric distribution:

1. The result of each draw (the elements of the population being sampled) can be classified into one of two mutually exclusive categories (e.g. Pass/Fail or Employed/Unemployed).
2. The successive draws are **not independent** of each other.
3. The number of draws is **fixed** in advance.
4. The probability of a success **changes** on each draw, as each draw decreases the population (*sampling without replacement* from a finite population).

Definition

A random variable X follows the hypergeometric distribution if its probability mass function (pmf) is given by

$$P(X = x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}, \quad x = 0, 1, 2, \dots, K \text{ if } K < n \text{ \& } x = 0, 1, 2, \dots, n \text{ if } n < K$$

where,

- N is the population size,
- K is the number of success in the population at the beginning of the experiment , - n is the number of draws (i.e. quantity drawn in each trial),
- k is the number of observed successes

Properties of hypergeometric distribution

1. It is discrete by nature.
2. It having range pf success $0, 1, 2, \dots, n \leq N$
3. It has three parameters n , k and N .
4. Mean and variance of hypergeometric distribution are

$$\text{Mean} = E(X) = n \frac{K}{N}$$

&

$$\text{Variance} = n \frac{K}{N} \frac{(N-K)}{N} \frac{N-n}{N-1}$$

5. In hypergeometric distribution mean > variance.
6. It based on sampling without replacement.
7. As $N \rightarrow \infty$ hypergeometric tends to binomial distribution.
8. Probability of success is not fixed in hypergeometric distribution.

Derivation of the Mean and Variance of the Hypergeometric Distribution

HYPERGEOMETRIC DISTRIBUTION

^{✓ 2018}
THEOREM 20:

SHOW THAT MEAN OF HYPERGEOMETRIC DISTRIBUTION IS

$$\mu = \frac{nK}{N}$$

Proof:

We know that
The pdf of hypergeometric distribution is

$$P(X=x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}, \quad 0 \leq x \leq n \text{ and } 0 \leq x \leq K$$

And

The mean μ is given as

$$\mu = E(X)$$

$$= \sum_{x=0}^n x \cdot P(X=x)$$

$$= \sum_{x=0}^n \frac{x \cdot \binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$$

$$= \sum_{x=0}^n x \frac{k!}{x!(k-x)!} \binom{N-k}{n-x}$$

$$= \sum_{x=0}^n \frac{x k! \binom{N-k}{n-x}}{x!(k-x)! \binom{N}{n}} = \sum_{x=1}^n \frac{k(k-1)! \binom{N-k}{n-x}}{(x-1)!(k-x)! \binom{N}{n}}$$

$$= \sum_{x=1}^n \frac{k(k-1)! \binom{N-k}{n-x}}{(x-1)!(k-x)! \binom{N}{n}} = \sum_{x=1}^n \frac{k \binom{k-1}{x-1} \binom{N-k}{n-x}}{\binom{N}{n}}$$

$$= \sum_{x=1}^n \frac{k \binom{k-1}{x-1} \binom{N-k}{n-x}}{\binom{N}{n}}$$

put $y = x-1$, $x = y+1$

$$\mu = \frac{k}{\binom{N}{n}} \sum_{y=0}^{n-1} \binom{k-1}{y} \binom{N-k}{n-y-1}$$

$$= \frac{k}{\binom{N}{n}} \binom{N-1}{n-1}$$

$$\left[\because \sum_{j=0}^m \binom{a}{j} \binom{b}{m-j} = \binom{a+b}{m} \right]$$

$$= \frac{k \binom{N-1}{n-1}}{\binom{N}{n}} = \frac{k n! (N-n)! (N-1)!}{N! (n-1)! (N-n)!}$$

$$\begin{aligned}
 & \frac{k n (n-1)}{N(N-1)!(n-1)!(N-n)!} \\
 &= \frac{k n (n-1)! (N-n)! (N-1)!}{N(N-1)!(n-1)!(N-n)!} \\
 &= \frac{k n}{N}
 \end{aligned}$$

$$\mu = \frac{n k}{N}$$

Hence proved

THEOREM 21: ²⁰¹⁸

SHOW THAT VARIANCE OF HPR.D IS

$$\sigma^2 = \frac{n k}{N} \left(\frac{N-k}{N} \right) \left(\frac{N-n}{N-1} \right)$$

Proof:

We know that

The pdf of HPR.D is

$$P(X=x) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}, \quad 0 \leq x \leq n$$

and

$$\begin{aligned}
 \sigma^2 &= E(X^2) - E(X)^2 \\
 E(X^2) &= \sum_{x=0}^n x^2 P(X=x)
 \end{aligned}$$

$$E(X^2) = \sum_{x=0}^n x^2 \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}} = \sum_{x=0}^n (x(x-1) + x) \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}$$

$$= \sum_{x=0}^n x(x-1) \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}} + \sum_{x=0}^n x \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}$$

we know that

$$E(X) = \sum_{x=0}^n x \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}} = \frac{n k}{N}$$

So

$$= \sum_{x=0}^n x(x-1) \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}} + \frac{n k}{N}$$

$$= \sum_{x=0}^n x(x-1) \frac{k!}{x!(k-x)!} \frac{\binom{N-k}{n-x}}{\binom{N}{n}} + \frac{n k}{N}$$

$$= \sum_{x=0}^n \frac{x(x-1) k! \binom{N-k}{n-x}}{x!(k-x)! \binom{N}{n}} + \frac{n k}{N}$$

$$= \sum_{x=2}^n \frac{x(x-1) k(k-1)(k-2)! \binom{N-k}{n-x}}{x(x-1)(x-2)!(k-x)! \binom{N}{n}} + \frac{n k}{N}$$

$$= \sum_{x=2}^n \frac{k(k-1)(k-2)! \binom{N-k}{n-x}}{(k-x)!(x-2)! \binom{N}{n}} + \frac{n k}{N}$$

$$= \sum_{k=2}^n \frac{k(k-1)(k-2)!}{(\cancel{k-2})(k-x)!} \frac{(N-k)}{(n-x)} + \frac{nK}{N}$$

$$= \sum_{k=2}^n \frac{k(k-1) \binom{N}{n} \binom{k-2}{\cancel{k-2}} \binom{N-k}{n-x}}{\binom{N}{n}} + \frac{nK}{N}$$

Let $y = x-2$, $x = y+2$

$$= \sum_{k=2}^n = \frac{k(k-1)}{\binom{N}{n}} \sum_{y=0}^{n-2} \binom{k-2}{y} \binom{N-k}{n-y-2} + \frac{nK}{N}$$

$$= \frac{k(k-1)}{\binom{N}{n}} \binom{N-2}{n-2} + \frac{nK}{N}$$

$$\left[\sum_{j=0}^m \binom{a}{j} \binom{b}{m-j} = \binom{a+b}{m} \right]$$

$$= \frac{k(k-1) \frac{(N-2)!}{(n-2)!(N-n)!}}{N!} + \frac{nK}{N}$$

$$\frac{n!(N-n)!}{n!(N-n)!}$$

$$= \frac{k(k-1)(N-2)! n!(N-n)!}{N!(n-2)!(N-n)!} + \frac{nK}{N}$$

$$= \frac{k(k-1) \cancel{(N-2)!} n(n-1) \cancel{(n-2)!} (N-n)!}{N(N-1) \cancel{(N-2)!} \cancel{(n-2)!} (N-n)!} + \frac{nK}{N}$$

=

$$= \frac{k(k-1)n(n-1)}{N(N-1)} + \frac{nk}{N}$$

$$\sigma^2 = E(X^2) - (E(X))^2$$

$$= \frac{k(k-1)n(n-1)}{N(N-1)} + \frac{nk}{N} - \left(\frac{nk}{N}\right)^2$$

$$= \frac{k(k-1)n(n-1)}{N(N-1)} + \frac{nk}{N} - \frac{n^2 k^2}{N^2}$$

$$= \frac{nk}{N} \left(\frac{(k-1)(n-1)}{N-1} + 1 - \frac{nk}{N} \right)$$

$$= \frac{nk}{N} \left(\frac{N(k-1)(n-1) + N(N-1) - (N-1)nk}{N(N-1)} \right)$$

$$= \frac{nk}{N} \left(\frac{n\cancel{k}N - Nn - \cancel{k}N + N + N^2 - \cancel{N} - N\cancel{n}k + nk}{N(N-1)} \right)$$

$$= \frac{nk}{N} \left(\frac{N^2 + nk - Nn - kN}{N(N-1)} \right)$$

$$= \frac{nk}{N} \left(\frac{N(N-n) - k(N-n)}{N(N-1)} \right)$$

$$= \frac{nk}{N} \left(\frac{(N-k)(N-n)}{N(N-1)} \right)$$

$$\sigma^2 = \frac{nK}{N} \left(\frac{N-K}{N} \right) \left(\frac{N-n}{N-1} \right)$$

Hence Proved

PROBLEMS OF HYPERGEOMETRIC DISTRIBUTION

Date: _____

Hypergeometric Distribution

Q No 1:

The names of 5 men and 5 women are written on slip of paper and placed in a hat. Four names are drawn. What is the probability that 2 are men and 2 are women?

Solution:

We are given that
 men = 5, women = 5, Drawn = 4 = n
 Total = N = 5 + 5 = 10, K = 5

We have to find
 $P(X=2) = ?$

As we know that
 the pdf of hypergeometric distribution is

$$P(X=x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}, \quad x=0,1,2,\dots,K \leq n$$

if $x=0,1,2,\dots,n \leq K$

$$P(X=2) = \frac{\binom{5}{2} \binom{10-5}{4-2}}{\binom{10}{4}} = \frac{(10)(10)}{210} = \frac{10}{21}$$

8.26

Let X denote the number of white beads. Then the probability that X assumes a value x , is

$$P(X=x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}} \quad \text{for } x=0,1,2,\dots,n$$

$K=0,1,2,\dots,K$

We are given that

$$K=4, \quad n=5, \quad N=11$$

$$x=0,1,2,3,4$$

x	$P(x)$	$xP(x)$	$x^2P(x)$
0	$\frac{\binom{4}{0} \binom{11-4}{5-0}}{\binom{11}{5}} = \frac{21}{462}$	0	0
1	$\frac{\binom{4}{1} \binom{11-4}{5-1}}{\binom{11}{5}} = \frac{140}{462}$	$\frac{140}{162}$	$\frac{140}{162}$
2	$\frac{\binom{4}{2} \binom{11-4}{5-2}}{\binom{11}{5}} = \frac{210}{462}$	$\frac{420}{462}$	$\frac{840}{462}$
3	$\frac{\binom{4}{3} \binom{11-4}{5-3}}{\binom{11}{5}} = \frac{84}{462}$	$\frac{252}{462}$	$\frac{756}{462}$
4	$\frac{\binom{4}{4} \binom{11-4}{5-4}}{\binom{11}{5}} = \frac{7}{462}$	$\frac{28}{462}$	$\frac{112}{462}$
$\frac{462}{462} = 1$		$\frac{840}{462}$	$\frac{1848}{462}$

$$\text{Mean} = \frac{\sum x P(x)}{\sum P(x)} = \frac{840}{462} = \frac{20}{11}$$

$$\begin{aligned} \text{Variance} &= \sum x^2 P(x) - (\sum x P(x))^2 \\ &= \frac{1842}{462} - \left(\frac{20}{11}\right)^2 \\ &= \frac{1842}{462} - \frac{400}{121} = 0.69179 \end{aligned}$$

By Formulas

$$\mu = \frac{nK}{N} = \frac{5(4)}{11} = \frac{20}{11}$$

$$\sigma^2 = \frac{nK}{N} \left(\frac{N-K}{N} \right) \left(\frac{N-n}{N-1} \right)$$

$$= \frac{20}{11} \left(\frac{11-4}{11} \right) \left(\frac{11-5}{11-1} \right)$$

$$= \frac{20}{11} \left(\frac{7}{11} \right) \left(\frac{6}{10} \right) = \frac{840}{1210} = 0.69$$

POISSON DISTRIBUTION

Poisson distribution is also known as rare event distribution. Poisson distribution is limiting form of binomial distribution under conditions

- ❖ “n” trials are sufficiently large
- ❖ “p” very small $\rightarrow 0$
- ❖ $0.05 \leq np \leq 15$ or $0.05 \leq \lambda \leq 15$

The probability distribution of Poisson distribution is

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad \text{for } x = 0, 1, 2, \dots, \infty$$

Properties of Poisson distribution

1. It is discrete by nature.
2. It having range $0, 1, 2, \dots, \infty$
3. It has only one parameters “ μ OR λ ”.
4. Mean and variance of Poisson distribution are same i.e. μ OR λ .
5. Its shape depends on its parameter μ OR λ .
6. Total area under the curve is unity.
7. Poisson distribution tends to normal if $n \rightarrow \infty$.

$$\beta_1 = 0; \quad \beta_2 = 3$$

8. It is +ve skewed distribution.
9. Its measure of skewness and kurtosis are

$$\beta_1 = \frac{1}{\mu}; \quad \beta_2 = 3 + \frac{1}{\mu}$$

10. If $X \sim p(u)$ & $Y \sim p(v)$ then,
 $X + Y \sim p(u + v)$ iff X & Y are independent
11. Probability generating function of Poisson distribution is

$$G(\theta) = e^{\mu(\theta-1)}$$

12. Moments generating function of Poisson distribution is

$$M_o(t) = e^{\mu(e^t-1)}$$

13. Characteristic function of Poisson distribution is

$$\theta(t) = e^{\mu(e^{it}-1)}$$

POISSON PROCESS

It is random physical mechanism in which event occur on a “time or distance scale” e.g. traffic death in a city per month, No. of typing errors per page, No. of persons born blind per year etc.

The pdf of Poisson process is

$$P(X = x) = \frac{e^{-\lambda t} (\lambda t)^x}{x!} \quad \text{for } x = 0, 1, 2, \dots, \lambda > 0$$

t = number of units of time

x = number of occurrences in t unit of time, and

λ = average number of occurrences per unit of time

Properties of Poisson process

1. Probability that an interval very short of time “ Δ ” in which an event occurs is $\Delta\mu$. “ μ ” is +ve quantity.
2. Probability that more than one event occur in a very short interval of time is negligible.
3. Probability of occurrence of event in disjoint time intervals is independent.

PROBLEM OF POISSON PROCESS

Date: _____

POISSON PROCESS

Qno1.

On the average, two minor accidents occur at a busy crossing - per day, what is the probability that, at the most two minor accidents will occur during an entire week?

Solution:

The occurrence rate of P.P = 2 per day = λ
one week = 7 days = t

Hence

$$\lambda t = 2 \times 7 = 14, \quad P(X \leq 2) = ?$$

as we know that

$$P(X=x) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}$$

$$P(X \leq 2) = P(X=0 \text{ or } X=1 \text{ or } X=2)$$

$$= \frac{e^{-14} (14)^0}{0!} + \frac{e^{-14} (14)^1}{1!} + \frac{e^{-14} (14)^2}{2!}$$

$$= e^{-14} + 14e^{-14} + 98e^{-14} = 113e^{-14}$$

$$= 0.00009396$$

DERIVATION OF MEAN AND VARIANCE OF POISSON DISTRIBUTION

2019 Show that the mean and the variance of Poisson distribution are equal (8 marks)

POISSON DISTRIBUTION

THEOREM 9:

Show that mean of poisson distribution is ' μ '.

Proof:

We know that pdf of poisson distribution is

$$P(X=x) = \frac{e^{-\mu} \mu^x}{x!}$$

$$x = 0, 1, 2, \dots, \infty$$

By definition of Expectation

$$E(X) = \sum_{x=0}^{\infty} x P(X=x)$$

$$= \sum_{x=0}^{\infty} x \frac{e^{-\mu} \mu^x}{x!} = \sum_{x=1}^{\infty} x \frac{e^{-\mu} \mu^x}{x(x-1)!}$$

$$= \sum_{x=1}^{\infty} \frac{e^{-\mu} \mu^{x-1} \mu}{(x-1)!} = e^{-\mu} \mu \sum_{x=1}^{\infty} \frac{\mu^{x-1}}{(x-1)!}$$

$$= e^{-\mu} \mu \left\{ \frac{\mu^0}{0!} + \frac{\mu^1}{1!} + \frac{\mu^2}{2!} + \dots + \infty \right\}$$

$$= e^{-\mu} \mu \{e^{\mu}\} = \mu \Rightarrow E(X) = \mu$$

Hence proved

THEOREM 11:

Show that variance of Poisson distribution is " μ "

Proof:

The pdf of poisson distribution is
$$P(X=x) = \frac{e^{-\mu} \mu^x}{x!}, \quad x=0, 1, 2, \dots, \infty$$

We know that

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \sum_{x=0}^{\infty} x^2 P(X=x)$$

$$= \sum_{x=0}^{\infty} x^2 \frac{e^{-\mu} \mu^x}{x!} = \sum_{x=0}^{\infty} (x(x-1) + x) \frac{e^{-\mu} \mu^x}{x!}$$

$$= \sum_{x=0}^{\infty} x(x-1) \frac{e^{-\mu} \mu^x}{x!} + \sum_{x=0}^{\infty} x \frac{e^{-\mu} \mu^x}{x!}$$

$$= \sum_{x=0}^{\infty} x(x-1) \frac{e^{-\mu} \mu^x}{x(x-1)(x-2)!} + \sum_{x=0}^{\infty} x \frac{e^{-\mu} \mu^x}{x(x-1)!}$$

$$= \sum_{x=0}^{\infty} \frac{e^{-\mu} \mu^{x-2} \mu^2}{(x-2)!} + \sum_{x=0}^{\infty} \frac{e^{-\mu} \mu^{x-1} \mu}{(x-1)!}$$

$$= e^{-\mu} \mu^2 \sum_{x=0}^{\infty} \frac{\mu^{x-2}}{(x-2)!} + e^{-\mu} \mu \sum_{x=0}^{\infty} \frac{\mu^{x-1}}{(x-1)!}$$

$$= e^{-\mu} \mu^2 \left\{ \frac{\mu^0}{0!} + \frac{\mu^1}{1!} + \frac{\mu^2}{2!} + \dots \right\} + e^{-\mu} \mu \left\{ \frac{\mu^0}{0!} + \frac{\mu^1}{1!} + \dots \right\}$$

$$= e^{-\mu} \mu^2 e^{\mu} + e^{-\mu} \mu e^{\mu}$$

$$= \mu^2 + \mu = \mu(\mu+1)$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

we know that

$$E(X) = \mu$$

$$\text{Var}(X) = \mu(\mu+1) - \mu^2$$

$$= \mu^2 + \mu - \mu^2 = \mu$$

$$\text{Var}(X) = \mu$$

+ hence proved

S.D

$$\text{S.D}(X) = \sqrt{\mu}$$

C.V

$$\text{C.V}(X) = \frac{\text{S.D}(X)}{\bar{X}} \times 100$$

$$= \frac{\sqrt{\mu}}{\sqrt{\mu}} \times 100 = \frac{1}{\sqrt{\mu}} \times 100$$

$$= \frac{100}{\sqrt{\mu}}$$

PGF & MGF OF POISSON DISTRIBUTION

PGF of Poisson Distribution

We know that

The pdf of poisson is

$$P(X=x) = \frac{e^{-u} u^x}{x!}, \quad x=0,1,2,\dots,\infty$$

Then

Probability Generating function is

$$E(\theta^x) = \sum_{x=0}^{\infty} \theta^x (P(X=x))$$

$$= \sum_{x=0}^{\infty} \theta^x \frac{e^{-u} u^x}{x!}$$

$$= \sum_{x=0}^{\infty} \frac{e^{-u} (\theta u)^x}{x!} = e^{-u} \sum_{x=0}^{\infty} \frac{(\theta u)^x}{x!}$$

$$= e^{-u} \left[1 + \frac{\theta u}{1!} + \frac{(\theta u)^2}{2!} + \dots + \infty \right]$$

$$= e^{-u} \cdot e^{\theta u} = e^{-u + \theta u} = e^{u(\theta - 1)}$$

$$E(\theta^x) = e^{u(\theta - 1)}$$

MGF of Poisson Distribution

To find moment generating function

put $\theta = e^t$

$$M(t) = E(e^{tx}) = e^{u(e^t - 1)}$$

moment generating function of poisson distribution is

$$M(t) = E(e^{tx}) = \sum_{x=0}^{\infty} e^{tx} \frac{e^{-u} u^x}{x!}$$

$$= \sum_{x=0}^{\infty} e^{-4} \frac{(e^t u)^x}{x!} = e^{-4} \sum_{x=0}^{\infty} \frac{(e^t u)^x}{x!}$$

$$= e^{-4} \left[1 + \frac{ue^t}{1!} + \frac{(ue^t)^2}{2!} + \dots + \infty \right]$$

$$M(t) = e^{-4} \{ e^{ue^t} \} = e^{-4 + ue^t} = e^{u(e^t - 1)} \rightarrow \textcircled{1}$$

differentiate $\textcircled{1}$ w.r.t t and applying limit $t \rightarrow 0$

$$\lim_{t \rightarrow 0} \frac{dM(t)}{dt} = \lim_{t \rightarrow 0} \frac{d}{dt} e^{u(e^t - 1)}$$

$$= \lim_{t \rightarrow 0} e^{u(e^t - 1)} u e^t \rightarrow \textcircled{2}$$

$$\mu_1' = e^{u(e^0 - 1)} u e^0 = u$$

$$\frac{d^2 M(t)}{dt^2} = u e^t e^{u(e^t - 1)} + u^2 e^{2t} e^{u(e^t - 1)}$$

$$\mu_2' = u e^0 e^{u(e^0 - 1)} + u^2 e^{2 \cdot 0} e^{u(e^0 - 1)}$$

$$\mu_2' = u + u^2$$

$$\mu_3' = u^3 + 3u^2 + u$$

$$\mu_4' = u^4 + 6u^3 + 7u^2 + u$$

Derivation of Poisson Approximation to the Binomial distribution

THEOREM 17

DERIVE THE POISSON DISTRIBUTION

Proof:

The poisson distribution is the limiting form of binomial distribution

as $n \rightarrow \infty$, $p \rightarrow 0$

The pdf of poisson distribution is
 $P(X=x) = \frac{e^{-\mu} \mu^x}{x!}$ for $x=0, 1, 2, \dots, \infty$

The pdf of binomial distribution is

$$P(X=x) = \binom{n}{x} p^x q^{n-x}, \quad x=0, 1, \dots, n \rightarrow \text{①}$$

we know

$$\mu = n p = n p \quad \text{if } n p = \mu, \quad p = \frac{\mu}{n}$$

$$q = 1 - p = 1 - \frac{\mu}{n}$$

putting in ①

$$\therefore \binom{n}{x} = \frac{n!}{x!(n-x)!}$$

$$P(X=x) = \binom{n}{x} \left(\frac{\mu}{n}\right)^x \left(1 - \frac{\mu}{n}\right)^{n-x}$$

$$= \frac{n!}{x!(n-x)!} \frac{\mu^x}{n^x} \left(1 - \frac{\mu}{n}\right)^n \left(1 - \frac{\mu}{n}\right)^{-x}$$

$$= \frac{n(n-1)(n-2)\dots(n-x+1)(n-x)!}{x! \cdot n \cdot n \cdot n \dots n(n-x)!} \mu^x \left(1 - \frac{\mu}{n}\right)^n \left(1 - \frac{\mu}{n}\right)^{-x}$$

$$= \frac{\mu^x \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{x-1}{n}\right) \left(1 - \frac{\mu}{n}\right)^n \left(1 - \frac{\mu}{n}\right)^{-x}}{x! \cdot \mu \cdot \mu \cdot \mu \dots \mu}$$

$$= \frac{\mu^x}{x!} \frac{\left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{x-1}{n}\right) \left(1 - \frac{\mu}{n}\right)^n}{\left(1 - \frac{\mu}{n}\right)^x}$$

Taking limit $n \rightarrow \infty$

$$= \frac{n^x}{x!} \frac{\left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{x-1}{n}\right)}{\left(1 - \frac{n}{n}\right)^x} \left[\lim_{n \rightarrow \infty} \left(1 - \frac{n}{n}\right)^{-\frac{n}{n}} \right]^{-1}$$

$$P(X=x) = \frac{n^x}{x!} e^{-n} \text{ OR } \frac{e^{-u} u^x}{x!}$$

$$x = 0, 1, 2, \dots, \infty$$

Hence

$$b(x; n, p) \Rightarrow P(X: n \text{ OR } u)$$

$$n \rightarrow \infty$$

MULTINOMIAL DISTRIBUTION

Multinomial Experiments

A binomial experiment has become multinomial experiment if there are more than two possible outcomes of each trial. Multinomial experiment has the following properties

- * Results of each trial classified into one of " k " mutually exclusive categories $C_1, C_2, \dots, C_k$
- * Successive trials are independent
- * Experiment is repeated a fixed no. of times say " n "
- * Probability of i th outcome is " p_i " which remains constant and $\sum_{i=1}^k p_i = 1$

Multinomial Distribution

A binomial distribution has become multinomial distribution if results of possible outcomes of an experiment exceeds 2 e.g.

manufactured item may be good; bad; average; & good
Road accident may caused; no injury; minor injury; severe injury; death; nothing etc.

Probability distribution of multinomial distribution

$$P(X=x_i) = \frac{n!}{\prod_{i=1}^k x_i!} \prod_{i=1}^k (p_i)^{x_i} \text{ where}$$

$$\sum_{i=1}^k p_i = 1 \text{ \& \& } \sum_{i=1}^k x_i = n$$

mean & variance of multinomial distribution are

$$E(x_i) = np_i$$

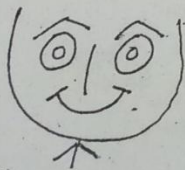
$$V(x_i) = np_i q_i$$

If $k=2$ multinomial distribution reduces to the binomial probability distribution.

Note Multinomial distribution in case of hyper

$$P(X=x_i) = \frac{\binom{n}{x_1} \binom{n-x_1}{x_2} \dots \binom{n-x_1-x_2-\dots-x_{k-1}}{x_k}}{\binom{n}{x_1, x_2, \dots, x_k}}$$

Musharraf-Ur-Rohman
Lecturer In Statistics
Govt Degree College Hfd



دنیا کی ہر چیز کا تریاق خود اسی میں مضمر ہے

عزت عورت کو شہسخت نہیں دے سکتی

Properties of multinomial distribution

Properties:

The mean of the Multinomial distribution

$$E(X_i) = np_i$$

The variance of the Multinomial distribution

$$\text{Var}(X_i) = np_i(1 - p_i)$$

The covariance of the Multinomial distribution

$$\text{cov}(X_i, X_j) = -np_i p_j, \quad i \neq j$$

NEGATIVE BINOMIAL DISTRIBUTION

-ve Binomial Distribution

In binomial experiment No. of success varies but No. of trial is fixed. But there are many experiments in which the no. of successes is fixed and No. of trials varies to produce fixed No. of successes. Such experiments are called -ve binomial experiment having following properties

- * Outcomes of each trial may be S or F
- * Probability of success fixed for all trials
- * Successive trials are independent
- * Experiment is repeated variable no. of times to obtain fixed no. of successes

If "X" denote No. of trials to produce K successes in -ve Binomial experiment it is called -ve binomial r.v. and its distribution is called -ve Binomial distribution with function

$$P(X=x) = \binom{x-1}{K-1} p^K q^{x-K} \quad x = K, K+1, K+2, \dots$$

It has 2 parameters K, $p > 0$

mean and variance of -ve Binomial distribution, are

$$E(X) = K/p \quad \text{Var}(X) = Kq/p^2$$

It is interesting to know that

Variance of -ve Binomial distribution is greater than its

Example:

Suppose that a game consists of tossing a fair coin until we obtain the 3rd head.

Then

head is ‘success’,

$$p=P(H)=0.5$$

&

$$r=3$$

In this case, the negative binomial distribution is given by

$$P(X = x) = \binom{x-1}{3-1} (0.5)^3 (1-0.5)^{x-3}, \quad x = 3, 3+1, 3+2, \dots$$

or

$$P(X = x) = \binom{x-1}{2} (0.5)^x, \quad x = 3, 4, 5, \dots$$

Calculation of Probabilities

x	$P(x)$
3	$P(X=3) = \binom{2}{2} (0.5)^3 = 0.1250$
4	$P(X=4) = \binom{3}{2} (0.5)^4 = 0.1875$
5	$P(X=5) = \binom{4}{2} (0.5)^5 = 0.1875$
6	$P(X=6) = \binom{5}{2} (0.5)^6 = 0.1562$
7	$P(X=7) = \binom{6}{2} (0.5)^7 = 0.1172$
8	$P(X=8) = \binom{7}{2} (0.5)^8 = 0.0820$
9	$P(X=9) = \binom{8}{2} (0.5)^9 = 0.0547$
10	$P(X=10) = \binom{9}{2} (0.5)^{10} = 0.0352$
And so on.	

Mean and Variance of the Negative Binomial Distribution

The mean of the distribution is given by

$$E(X) = \frac{r(1-p)}{p}$$

and the Variance is

$$\text{Var}(X) = \frac{r(1-p)}{p^2}$$

MGF of the Negative Binomial Distribution

The moment generating function of the distribution is given by

$$M(t) = \left(\frac{1-p}{1-pe^t} \right)^r \text{ for } t < -\log p$$

This can be used to find the mean, variance and higher moments of the distribution.

GEOMETRIC DISTRIBUTION

Assume Bernoulli trials — that is,

- (1) there are two possible outcomes,
- (2) the trials are independent, and
- (3) p , the probability of success, remains the same from trial to trial.

(4) n , the number of trials is fixed in advance
and

the binomial distribution is given by

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, 3, \dots, n$$

When $n=1$,
there are two possible outcomes,
and p , the probability of success, is the only parameter of
the Bernoulli distribution given by

$$p_X(x) = \binom{1}{x} p^x (1-p)^{1-x}, \quad x = 0, 1$$

or

$$p_X(x) = p^x (1-p)^{1-x}, \quad x = 0, 1$$

If we let X denote the number of trials **until the first success**.

Then, the **probability mass function** of X is:

$$p_X(x) = (1-p)^{x-1} p, \quad x = 1, 2, 3, \dots$$

Date: _____

GEOMETRIC DISTRIBUTION

Q no 1

Suppose that a game consists of tossing a fair coin until we obtain the first head

$$\text{Then } p = P(H) = \frac{1}{2}$$

And, in this G.D is given by

$$P_x(x) = (1-p)^{x-1} p, \quad x = 1, 2, 3, \dots$$

$$P_x(x) = \left(1 - \frac{1}{2}\right)^{x-1} \left(\frac{1}{2}\right)$$

$$P_x(x) = \left(\frac{1}{2}\right)^{x-1} \left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^{x-1+1} = \left(\frac{1}{2}\right)^x, \quad x = 1, 2, 3, \dots$$

$$P_x(x) = \frac{1}{2}, \quad x = 1, 2, 3, \dots$$

Representation in tabular form

X	P(x)
1	$\left(\frac{1}{2}\right)^1 = 0.5$
2	$\left(\frac{1}{2}\right)^2 = 0.25$
3	$\left(\frac{1}{2}\right)^3 = 0.125$
4	$\left(\frac{1}{2}\right)^4 = 0.0625$
5	$\left(\frac{1}{2}\right)^5 = 0.03125$
6	$\left(\frac{1}{2}\right)^6 = 0.015625$

GEMMA DISTRIBUTION

Formal definition:

Gamma distribution

$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} & 0 < x < \infty, \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases}$$

is said to be the pdf of a gamma distribution with shape parameter α and scale parameter β and we write this as X has a $\Gamma(\alpha, \beta)$ distribution.

Shape of gamma distribution

- From the graph it is obvious that, for any fixed value of the scale parameter β , as **α increases**, the shape of the distribution tends to normality.

First we introduce the **gamma (Γ) function**:

In calculus, the integral

$$\int_0^{\infty} y^{\alpha-1} e^{-y} dy$$

exists for $\alpha > 0$ and that the value of the integral is a positive number.

The integral is called the **gamma function**, and we write

$$\Gamma(\alpha) = \int_0^{\infty} y^{\alpha-1} e^{-y} dy$$

Properties of gamma distribution

Properties of the Gamma Function

- If $\alpha = 1$, clearly (easy to verify through integration)
$$\Gamma(1) = \int_0^{\infty} e^{-y} dy = 1$$
- If $\alpha > 1$, an integration by parts show that

$$\Gamma(\alpha) = (\alpha - 1) \int_0^{\infty} y^{\alpha-2} e^{-y} dy = (\alpha - 1) \Gamma(\alpha - 1).$$

- Accordingly, if α is a positive **integer** greater than 1,

$$\Gamma(\alpha) = (\alpha - 1)(\alpha - 2) \cdots (3)(2)(1) \Gamma(1) = (\alpha - 1)!.$$

- Since $\Gamma(1) = 1$, this suggests we take **$0! = 1$** .
- It is important to note that **the value of a gamma function is always positive**.

CDF of the Gamma Distribution

The CDF of the distribution is given by

$$F(x) = \begin{cases} \frac{1}{\Gamma(\alpha)} \gamma\left(\alpha, \frac{x}{\beta}\right) & \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where this $\gamma\left(\alpha, \frac{x}{\beta}\right)$ is the incomplete gamma function.

INCOMPLETE GAMMA FUNCTION

The upper incomplete gamma function is defined as:

$$\Gamma(\alpha, y) = \int_y^{\infty} t^{\alpha-1} e^{-t} dt,$$

whereas the lower incomplete gamma function is defined as:

$$\gamma(\alpha, y) = \int_0^y t^{\alpha-1} e^{-t} dt.$$

It is interesting to note that $\gamma(\alpha, y) + \Gamma(\alpha, y) = \Gamma(\alpha)$.

MEAN AND VARIANCE OF GAMMA DISTRIBUTION

Mean: $\mu_X = E(X) = \alpha\beta$

Variance: $\sigma_X^2 = \alpha\beta^2$

MGF OF GAMMA DISTRIBUTION

The moment-generating function of the gamma distribution is

$$M(t) = (1 - \beta t)^{-\alpha} \text{ for } t < \frac{1}{\beta}$$

Find the value of $\Gamma(5/2) = ?$
as we know that $\Gamma(5/2) = ?$

$$\Gamma(\alpha) = \int_0^{\infty} y^{\alpha-1} e^{-y} dy \rightarrow (1)$$

This can be further simplified as follow:

$$\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1) \rightarrow (2)$$

By using (2)

$$\begin{aligned}\Gamma(5/2) &= (\alpha-1)\Gamma(\alpha-1) \\ &= (5/2-1)\Gamma(5/2-1) \\ &= (3/2)\Gamma(3/2)\end{aligned}$$

Now by calculating

$$\Gamma(3/2)$$

$$\text{using } \Gamma(n+1) = n\Gamma(n)$$

$$\therefore \Gamma(1/2) = \sqrt{\pi}$$

$$\Gamma(n) = \frac{1}{n}\Gamma(n+1)$$

$$n = 3/2$$

$$\Gamma(3/2) = \Gamma(1/2+1)$$

$$= \frac{1}{2}\Gamma(1/2) = \frac{1}{2}\sqrt{\pi}$$

$$\Gamma(5/2) = \frac{3}{2}\Gamma(3/2) = \frac{3}{2}\left(\frac{1}{2}\sqrt{\pi}\right)$$

$$\Gamma(5/2) = \frac{3}{4}\sqrt{\pi} = \left(\frac{3}{4}\pi\right)^{1/2}$$

Beta Distribution

The **probability density function** of the β distribution is

$$f(x; \alpha, \beta) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 \leq x \leq 1; \alpha, \beta > 0$$

where,

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

and α and β are the **shape parameters** respectively.

About the Beta function B

- The Beta function B in the denominator plays the role of a “**normalizing constant**” which assures that the total area under the density curve equals 1.
- The Beta function is equal to a **ratio** of Gamma functions:

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

- Keeping in mind that for integers, $\Gamma(k) = (k-1)!$, one can do some checking and get an idea of what the shape might be.

Cumulative distribution Function

- The formula for the cumulative distribution function of the beta distribution is also called the **incomplete β function ratio** (commonly denoted by I_x) and is defined as

$$F(x) = I_x(\alpha, \beta) = \frac{\int_0^x t^{\alpha-1} (1-t)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 \leq x \leq 1; \alpha, \beta > 0$$

where B is the beta function defined above.

Properties

Some Properties:

- Mean** $= \mu$

$$= E(x) = \alpha / (\alpha + \beta)$$

- Variance**(x)

$$= \alpha\beta / (\alpha + \beta)^2 (\alpha + \beta + 1)$$

Mode:

- If $\alpha > 1$ and $\beta > 1$, the peak of the density is in the interior of $[0,1]$ and mode of the Beta distribution is

$$\text{mode} = \alpha - 1 / \alpha + \beta - 2$$

If α or $\beta < 1$, the mode may be at an edge.

PROBLEM OF BETA DISTRIBUTION

Date: _____

BETA DISTRIBUTION

Qno1 Suppose that the proportion of defective DVDs in a large shipment follows a Beta distribution with $\alpha=2$ and $\beta=5$. What is the probability that the shipment has 20% to 30% defective DVDs?

Solution:

Let x represent the proportion of defective DVDs in this large shipment.

The probability density function of the Beta distribution is given by

$$f(x) = \frac{x^{\alpha-1} (1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 \leq x \leq 1, \alpha > 0, \beta > 0$$

and we are given that

$$\alpha = 2, \beta = 5$$

so

$$f(x) = \frac{x^{2-1} (1-x)^{5-1}}{B(2, 5)}, \quad 0 \leq x \leq 1$$

Date: _____

we have to find that

$$P(20\% \leq x \leq 30\%) = P(0.20 \leq x \leq 0.30)$$

and

$$P(0.20 \leq x \leq 0.30) = \int_{0.20}^{0.30} \frac{x^{2-1} (1-x)^{5-1}}{B(2,5)} dx$$

First of all we solve the $B(2,5)$

as we know that

$$B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

Here

$$B(2,5) = \frac{\Gamma(2)\Gamma(5)}{\Gamma(2+5)} = \frac{\Gamma(2)\Gamma(5)}{\Gamma(7)}$$

$$= \frac{(2-1)!(5-1)!}{(7-1)!} = \frac{(1)!(4)!}{(6)!}$$

$$B(2,5) = 0.03333$$

Date: _____

Now

$$P(0.20 \leq x \leq 0.30) = \int_{0.20}^{0.30} \frac{x(1-x)^4}{0.033333} dx$$

$$= \frac{1}{0.033333} \int_{0.2}^{0.3} x(1-x)^4 dx$$

By solving this we must get

$$P(0.2 \leq x \leq 0.3) = 0.2352$$

NORMAL DISTRIBUTION

Definition:

A random variable X is said to have a normal distribution if its pdf is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty < x < \infty.$$

- μ is the mean or expectation of the distribution (and also its median and mode),
- $\sigma > 0$ is the standard deviation, and
- σ^2 is the variance.

A random variable X is said to have a normal distribution if its pdf is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$

where μ is the mean of the distribution and $\sigma > 0$ is the standard deviation.

We say that X has the $N(\mu, \sigma^2)$ distribution.

1. For this purpose, we let $Z = \frac{X - \mu}{\sigma}$.

Then, it is easy to prove that $Z \sim N(0,1)$.

PROPERTIES OF NORMAL DISTRIBUTION

The following are the main properties of the normal distribution:

1- Continuous Distribution

The normal probability distribution is a continuous distribution that ranges from $-\infty$ to $+\infty$.

2- Total Probability

The total area under the normal curve is unity.

3- Mode and Maximum ordinate

The normal probability function is unimodal, its mode is μ and its maximum ordinate at $x = \mu$ is

$$f(\mu) = \frac{1}{\sigma\sqrt{2\pi}}$$

4- Symmetrical Distribution

The normal probability distribution is a symmetrical distribution. Thus

i) Mean = Median = Mode

ii) $x_{0.75} - \mu = \mu - x_{0.25}$ Where, μ is mean, $x_{0.25}$ is 1st Quartile and $x_{0.75}$ is 3rd Quartile.

iii) All odd order moments about mean are zero.

$$\mu_1 = \mu_3 = \mu_5 = \dots = 0$$

5- Special areas under the curve

i) $P(\mu - \sigma < X < \mu + \sigma) = 0.6827$

ii) $P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.9545$

iii) $P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.9973$

NOTE:

yh total 11 properties hain, marks and time k according kr lyna jo easy lagan

6- Median, Quartiles, Quartile Deviation

$$x_{0.5} = \mu, \quad x_{0.25} = \mu - 0.6745\sigma, \quad x_{0.75} = \mu + 0.6745\sigma$$

$$Q.D(X) = \frac{x_{0.75} - x_{0.25}}{2} = 0.6745\sigma \approx \frac{2}{3}\sigma$$

$$\mu = \frac{x_{0.25} + x_{0.75}}{2}, \quad \sigma = \frac{x_{0.75} - x_{0.25}}{1.349}$$

7- Variance, Standard Deviation and Mean Deviation

$$Var(X) = \sigma^2$$

$$S.D(X) = \sigma$$

$$M.D(X) = \sigma \sqrt{\frac{2}{\pi}} = 0.7979\sigma \approx \frac{4}{5}\sigma$$

8- Moments and Moment Ratios

$$\mu_1 = 0, \quad \mu_2 = \sigma^2, \quad \mu_3 = 0, \quad \mu_4 = 3\sigma^4$$

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{0}{(\sigma^2)^3} = 0, \quad \beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{3\sigma^4}{(\sigma^2)^2} = 3$$

9- Point of Inflexion

The points of inflexion of the normal probability density function are equidistant from mean μ , are at $x = \mu - \sigma$ and $x = \mu + \sigma$. The normal distribution is a bell shaped distribution.

10- Asymptotic Curve

The normal curve is asymptotic to the x- axis, that is, $|x|$ grows without bound, the curve get closer and closer to the x- axis but always stay above it.

11- Reproductive Property

If X_1 and X_2 are two independent normal random variables having distributions $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$

respectively, their sum $X_1 + X_2$ is also a normal variable having the distribution.

$$N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

STANDARD NORMAL DISTRIBUTION

- The standard normal distribution is a **special** case of the normal distribution.
- It is the distribution that occurs when a normal random variable has a **mean** of zero and a **standard deviation** of one.

In other words, if the random variable Z follows a normal distribution with parameters $(0, 1)$, i.e. $Z \sim N(0, 1)$ then we say Z is a standard normal random variable.

Date: _____ NORMAL DISTRIBUTION

Qnd let X be $N(2, 25)$ and suppose that we are interested in computing the probability $P(0 < X < 10)$

Solution:

we are given that $N(2, 25)$
Here $\mu = 2$, $\sigma^2 = 25$, $\sigma = \sqrt{25} = 5$
and, we have to find

$P(0 < X < 10) = ?$

as we know that

$$P(0 < X < 10) = F(10) - F(0)$$
$$= \Phi\left(\frac{10-2}{5}\right) - \Phi\left(\frac{0-2}{5}\right)$$
$$= \Phi\left(\frac{8}{5}\right) - \Phi\left(-\frac{2}{5}\right)$$
$$= \Phi(1.6) - \Phi(-0.4)$$
$$= 0.94520 - 0.34458$$
$$= 0.60062 \approx 60\%$$

$\therefore Z = \frac{X - \mu}{\sigma}$

Table 9

Values of the Standard Normal Cumulative Distribution Function

Tabulated Values: $\Phi(z) = P(Z \leq z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-4.5	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
-4.4	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001
-4.3	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001
-4.2	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001
-4.1	0.00002	0.00002	0.00002	0.00002	0.00002	0.00002	0.00002	0.00002	0.00001	0.00001
-4.0	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	0.00002	0.00002	0.00002	0.00002
-3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
-3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
-3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
-3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
-3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
-3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
-3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
-3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
-3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
-3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00104	0.00100
-2.9	0.00187	0.00181	0.00175	0.00169	0.00164	0.00159	0.00154	0.00149	0.00144	0.00139
-2.8	0.00256	0.00248	0.00240	0.00233	0.00226	0.00219	0.00212	0.00205	0.00199	0.00193
-2.7	0.00347	0.00336	0.00326	0.00317	0.00307	0.00298	0.00289	0.00280	0.00272	0.00264
-2.6	0.00466	0.00453	0.00440	0.00427	0.00415	0.00402	0.00391	0.00379	0.00368	0.00357
-2.5	0.00621	0.00604	0.00587	0.00570	0.00554	0.00539	0.00523	0.00508	0.00494	0.00480
-2.4	0.00820	0.00798	0.00776	0.00755	0.00734	0.00714	0.00695	0.00676	0.00657	0.00639
-2.3	0.01072	0.01044	0.01017	0.00990	0.00964	0.00939	0.00914	0.00889	0.00866	0.00842
-2.2	0.01390	0.01355	0.01321	0.01287	0.01255	0.01222	0.01191	0.01160	0.01130	0.01101
-2.1	0.01786	0.01743	0.01700	0.01659	0.01618	0.01578	0.01539	0.01500	0.01463	0.01426
-2.0	0.02275	0.02222	0.02169	0.02118	0.02068	0.02018	0.01970	0.01923	0.01876	0.01831
-1.9	0.02872	0.02807	0.02743	0.02680	0.02619	0.02559	0.02500	0.02442	0.02385	0.02330
-1.8	0.03593	0.03515	0.03438	0.03362	0.03288	0.03216	0.03144	0.03074	0.03005	0.02938
-1.7	0.04457	0.04363	0.04272	0.04182	0.04093	0.04006	0.03920	0.03836	0.03754	0.03673
-1.6	0.05480	0.05370	0.05262	0.05155	0.05050	0.04947	0.04846	0.04746	0.04648	0.04551
-1.5	0.06681	0.06552	0.06426	0.06301	0.06178	0.06057	0.05938	0.05821	0.05705	0.05592

Table 9 (Continued)

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-1.4	0.08076	0.07927	0.07780	0.07636	0.07493	0.07353	0.07215	0.07078	0.06944	0.06811
-1.3	0.09680	0.09510	0.09342	0.09176	0.09012	0.08851	0.08692	0.08534	0.08379	0.08226
-1.2	0.11507	0.11314	0.11123	0.10935	0.10749	0.10565	0.10383	0.10204	0.10027	0.09853
-1.1	0.13567	0.13350	0.13136	0.12924	0.12714	0.12507	0.12302	0.12100	0.11900	0.11702
-1.0	0.15866	0.15625	0.15386	0.15151	0.14917	0.14686	0.14457	0.14231	0.14007	0.13786
-0.9	0.18406	0.18141	0.17879	0.17619	0.17361	0.17106	0.16853	0.16602	0.16354	0.16109
-0.8	0.21186	0.20897	0.20611	0.20327	0.20045	0.19766	0.19489	0.19215	0.18943	0.18673
-0.7	0.24196	0.23885	0.23576	0.23270	0.22965	0.22663	0.22363	0.22065	0.21770	0.21476
-0.6	0.27425	0.27093	0.26763	0.26435	0.26109	0.25785	0.25463	0.25143	0.24825	0.24510
-0.5	0.30854	0.30503	0.30153	0.29806	0.29460	0.29116	0.28774	0.28434	0.28096	0.27760
-0.4	0.34458	0.34090	0.33724	0.33360	0.32997	0.32636	0.32276	0.31918	0.31561	0.31207
-0.3	0.38209	0.37828	0.37448	0.37070	0.36693	0.36317	0.35942	0.35569	0.35197	0.34827
-0.2	0.42074	0.41683	0.41294	0.40905	0.40517	0.40129	0.39743	0.39358	0.38974	0.38591
-0.1	0.46017	0.45620	0.45224	0.44828	0.44433	0.44038	0.43644	0.43251	0.42858	0.42465
-0.0	0.50000	0.49601	0.49202	0.48803	0.48405	0.48006	0.47608	0.47210	0.46812	0.46414
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91308	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96994	0.97062
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670

Table 9 (Continued)

[illegible]

Q No 1

What is BMI?

BMI (Body mass Index) is a person's weight in kilograms divided by the square of height in meters.

A high BMI can be an indicator of high body fatness.

Q No 2

The BMI for men aged 60 is normally distributed with the mean equal to 29 with a S.D of 6 whereas the BMI for women aged 60 is normally distributed with mean 28 and S.D 7.

What is the 90th percentile of BMI for men?

Solution:

We are given that

$$\mu_m = 29, \quad \sigma_m = 6$$

$$\mu_w = 28, \quad \sigma_w = 7$$

We have to interested to find the 90th percentile of BMI for men.

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We have found $z = 1.28$ for percentile 90th

As we know that

$$z = \frac{x - \mu}{\sigma}$$

$$z\sigma = x - \mu$$

$$x = \mu + z\sigma$$

$$x = 29 + 6(1.28)$$

$$= 36.68$$

Table 10 (b)

Specific Quantiles of the Standard Normal Distribution

Tabulated Values: $z_p = \Phi^{-1}(p)$

$$\Phi(z_p) = P(Z \leq z_p) = \int_{-\infty}^{z_p} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = p$$

p	0.0005	0.001	0.005	0.01	0.025	0.05	0.10	0.20	0.30	0.40
z_p	-3.291	-3.090	-2.576	-2.326	-1.960	-1.645	-1.282	-0.842	-0.524	-0.253
p	0.60	0.70	0.80	0.90	0.95	0.975	0.990	0.995	0.999	0.9995
z_p	0.253	0.524	0.842	1.282	1.645	1.960	2.326	2.576	3.090	3.291

EXPONENTIAL DISTRIBUTION

The Probability density function (pdf) of an exponential distribution is

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \lambda > 0 \\ 0 & x < 0. \end{cases}$$

So it is obvious that is a **single-parameter** distribution.

Let $\theta = \frac{1}{\lambda}$

then,

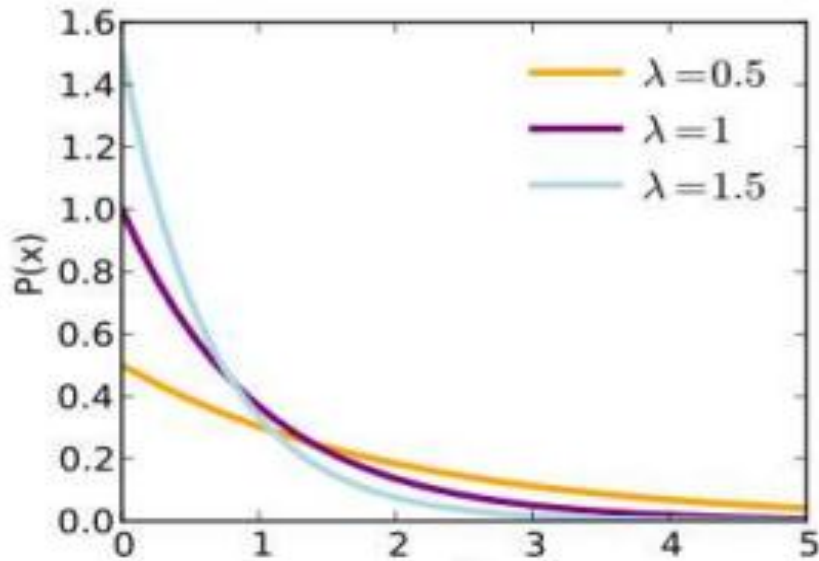
mean = θ

and

$$f(x; \theta) = \begin{cases} \frac{1}{\theta} e^{-\frac{x}{\theta}} & x \geq 0, \theta > 0 \\ 0 & x < 0. \end{cases}$$

So it is obvious that is a **single-parameter** distribution.

SHAPE OF EXPONENTIAL DISTRIBUTION



Cumulative distribution function (cdf) of exponential distribution

The Cumulative distribution function (cdf) of an exponential distribution is

$$F(x; \lambda) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0, \lambda > 0 \\ 0 & x < 0. \end{cases}$$

The Cumulative distribution function (cdf) of an exponential distribution in terms of θ

$$F(x; \theta) = \begin{cases} 1 - e^{-\frac{x}{\theta}} & x \geq 0, \theta > 0 \\ 0 & x < 0. \end{cases}$$

Shape of Cumulative distribution function (cdf) of an exponential distribution

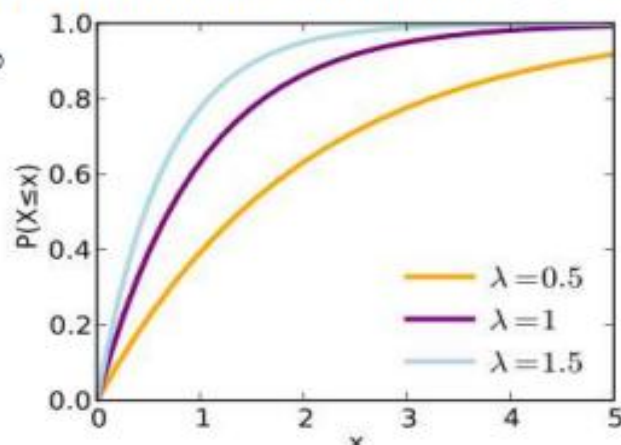
CDF of the **Exponential Distribution** with mean=0.5, 1, 1.5

$$F(x) = 1 - e^{-x}, \quad 0 < x < \infty$$

$$\text{--- } F(x) = 1 - e^{-\frac{x}{0.5}}, \quad 0 < x < \infty$$

$$\text{--- } F(x) = 1 - e^{-x}, \quad 0 < x < \infty$$

$$\text{--- } F(x) = 1 - e^{-\frac{x}{1.5}}, \quad 0 < x < \infty$$



MEAN AND STANDARD DEVIATION OF EXPONENTIAL DISTRIBUTION

MEAN

$$E(X) \text{ or } \mu = \frac{1}{\lambda}.$$

STANDARD DEVIATION

The **mean and standard deviation** of the exponential distribution are equal.

PROBLEMS OF EXPONENTIAL DISTRIBUTION

① we are given that
 $\lambda = 1$
we have to find
 $P(X > 3)$
as we know that $\Rightarrow f(x) = \lambda e^{-\lambda x}$
$$P(X > 3) = \int_3^{\infty} (1) e^{-x} dx$$
$$= \left[-e^{-x} \right]_3^{\infty} = \left[-e^{-\infty} - (-e^{-3}) \right]$$
$$= 0 + e^{-3} = e^{-3}$$
$$\boxed{P(X > 3) = e^{-3}}$$

$f(x) = 3e^{-3x}$ for $x > 0$
Find probability that x is not less than 2?
$$P(X > 2) = \int_2^{\infty} 3e^{-3x} dx$$
$$= -\frac{3}{3} \left[e^{-3x} \right]_2^{\infty}$$
$$= -1 \left[e^{-3(\infty)} - e^{-3(2)} \right]$$
$$= -1 (0 - e^{-6})$$
$$\boxed{P(X > 2) = e^{-6}}$$

$$f(x) = \frac{1}{3} e^{-x/3}$$

$$P(X > 5) = ?$$

$$P(X > 5) = \int_5^{\infty} \frac{1}{3} e^{-x/3} dx$$

$$= \frac{-3}{3} \left[e^{-x/3} \right]_5^{\infty}$$

$$= -1 \left[e^{-\infty/3} - e^{-5/3} \right]$$

$$= -1 (0 - e^{-5/3})$$

$$\boxed{P(X > 5) = e^{-5/3}}$$

the MGF of the Exponential Distribution is given

$$M_0(t) = \frac{\lambda}{\lambda - t} \quad \text{for } t < \lambda.$$

UNIFORM DISTRIBUTION

There are two types of uniform distribution:

1. Discrete uniform distribution
2. Continuous uniform distribution

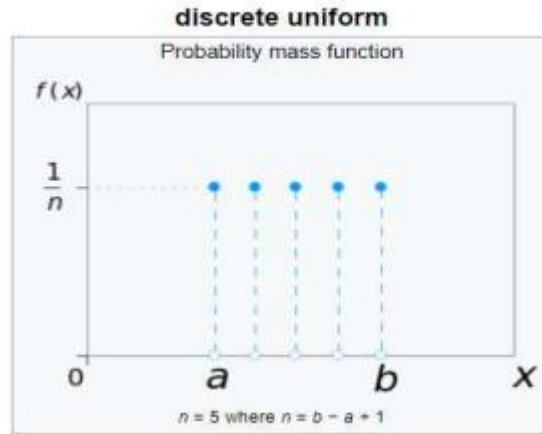
DISCRETE UNIFORM DISTRIBUTION

The **discrete uniform distribution** is a **symmetric** probability distribution whereby a finite number of values are **equally likely** to be observed;

In other words, if the discrete random variable X can assume n distinct values, and every one of the n values has **equal probability** $1/n$, we are dealing with a discrete uniform distribution.

SHAPE OF THE DISCRETE UNIFORM DISTRIBUTION

SHAPE OF THE DISTRIBUTION



MEAN AND STANDARD DEVIATION OF DISCRETE UNIFORM DISTRIBUTION

MEAN

$$\mu = E(X) = \frac{n+1}{2}.$$

STANDARD DEVIATION

$$\sigma = \sqrt{\frac{n^2 - 1}{12}}$$

EXAMPLES OF DISCRETE UNIFORM DISTRIBUTION

Example 1:

Let X denote the number of heads that we can have if a fair coin is tossed once. Then, we have

X	$p(x)$
0	$\frac{1}{2}$
1	$\frac{1}{2}$

Example 2:

Let X denote the number that we observe on the uppermost face if we roll a fair die once.

SOLUTION

X	$p(x)$
1	$\frac{1}{6}$
2	$\frac{1}{6}$
3	$\frac{1}{6}$
4	$\frac{1}{6}$
5	$\frac{1}{6}$
6	$\frac{1}{6}$

Cumulative Distribution Function:

Step 1: Find the cumulative probabilities:

X	$p(x)$	<i>cumulative</i>
1	1/6	1/6
2	1/6	2/6
3	1/6	3/6
4	1/6	4/6
5	1/6	5/6
6	1/6	6/6

Step 2: Write as follows:

$$p(x) = \begin{cases} 0 & -\infty < x < 1 \\ 1/6 & 1 \leq x < 2 \\ 2/6 & 2 \leq x < 3 \\ 3/6 & 3 \leq x < 4 \\ 4/6 & 4 \leq x < 5 \\ 5/6 & 5 \leq x < 6 \\ 6/6 & 6 \leq x < \infty \end{cases}$$

The entire X -axis (from $-\infty$ to ∞) must be covered.

PARAMETER

The constants a and b are called the *parameters* of the discrete uniform distribution.

MGF

$$M(t) = \frac{e^t - e^{(n+1)t}}{n(1 - e^t)}.$$

CONTINUOUS UNIFORM DISTRIBUTION

Probability Distribution Function

The probability density function of the continuous uniform distribution is:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b, \\ 0 & \text{for } x < a \text{ or } x > b \end{cases}$$

Cumulative Distribution Function

The Cumulative Distribution function of the continuous uniform distribution is:

$$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b, \\ 1 & \text{for } x > b \end{cases}$$

Expectation and Variance

If $X \sim U(a, b)$, then:

$$E(X) = \frac{(a+b)}{2}$$

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

CUMULANT GENERATING FUNCTION

Definition of Cumulant Generating Function

For any distribution for which the mgf exists, the cumulant-generating function $K(t)$ is the natural logarithm of the moment generating function i.e.

$$K(t) = \ln E(e^{tX})$$

For the **normal distribution** with expected value μ and variance σ^2 , the cumulant generating function is

$$K(t) = \mu t + \frac{\sigma^2 t^2}{2}.$$

Thus, in a nutshell, the cumulants of the normal distribution with expected value μ and variance σ^2 are

$$K_1 = \mu,$$

$$K_2 = \sigma^2,$$

and

$$K_3 = K_4 = \dots = 0.$$

all the higher cumulants will be zero.

BIVARIATE NORMAL DISTRIBUTION

Definition

A pair of random variables X_1 and X_2 have a bivariate normal distribution and they are referred to as **jointly normally distributed** random variables **if and only if** their joint probability density is given by

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{\left\{ -\frac{1}{2(1-\rho^2)} \left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho \left(\frac{x_1-\mu_1}{\sigma_1} \right) \left(\frac{x_2-\mu_2}{\sigma_2} \right) + \frac{(x_2-\mu_2)^2}{\sigma_2^2} \right) \right\}}$$

for $-\infty < x_1 < \infty$ and $-\infty < x_2 < \infty$,

where $\mu_1 \in \mathbb{R}$, $\mu_2 \in \mathbb{R}$, $\sigma_1 > 0$, $\sigma_2 > 0$, and $-1 \leq \rho \leq 1$.

What is meant by the 'X₁X₂ floor'?

Consider the three-dimensional space for example the room in which you are sitting right now.

- Now focus on the floor of the room.
- One 'edge' of the floor will be called the X_1 axis
- The other 'edge' of the floor will be called the X_2 axis

And, as such, the floor will be called the ' X_1X_2 floor'
(actually it is the X_1X_2 -plane)

What is meant by contour?

By the term '**contour**', we mean the **set of points on the X_1X_2 floor** corresponding to which the **ordinates are of equal height**.

Definition of Independence:

Two **random** variables X and Y are **independent**, **statistically independent** or **stochastically independent** if

$$f_{XY}(x, y) = f_X(x) f_Y(y) \quad \dots(1)$$

i.e.

if the **joint** density $f_{XY}(x, y)$ is equal to the product of the **marginal** density $f_X(x)$ and the **marginal** density $f_Y(y)$.

DISCRETE DISTRIBUTION

Distribution	Moment Generating Function (MGF)
Bernoulli $P(X=1) = p$	$1 - p + pe^t$
Geometric $(1-p)^{k-1}p$	$\frac{pe^t}{1-(1-p)e^t} \forall t < -\ln(1-p)$
Binomial $B(n, p)$	$(1 - p + pe^t)^n$
Poisson λ	$e^{\lambda(e^t - 1)}$
Negative binomial $NB(r, p)$	$\frac{(1-p)^r}{(1-pe^t)^r}$
Uniform (discrete) $U(a, b)$	$\frac{e^{at} - e^{(b+1)t}}{(b-a+1)(1-e^t)}$

CONTINUOUS DISTRIBUTION

Distribution	Moment Generating Function (MGF)
Uniform (continuous) $U(a, b)$	$\frac{e^{tb} - e^{ta}}{t(b-a)}$
Normal $N(\mu, \sigma^2)$	$e^{t\mu + \frac{1}{2}\sigma^2 t^2}$
Chi-Squared χ^2_k	$(1 - 2t)^{-\frac{k}{2}}$
Gamma $\Gamma(k, \theta)$	$(1 - t\theta)^{-k}; \forall t < \frac{1}{\theta}$
Exponential λ	$(1 - t\lambda^{-1})^{-1}, (t < \lambda)$
Laplace $L(\mu, b)$	$\frac{e^{t\mu}}{1 - b^2 t^2}$
Cauchy $N(\mu, \theta)$	Do not exist

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* Important abbreviation

PMF = Probability mass function

PDF = Probability Density function

CDF = Cumulative distribution function

MGF = Moment generating function

CGF = Cumulative generating function

* Parameters

Discrete uniform distribution (a, b)

Binomial distribution (n, p)

Gamma distribution (α, β)

Beta distribution (α, β)

Normal distribution (μ, σ)

* Discrete uniform distribution

PMF
$$P(X) = \begin{cases} \frac{1}{n} & x = 0, 1, 2, \dots \\ 0 & \text{elsewhere} \end{cases}$$

Mean
$$\mu = \frac{n+1}{2}$$

Variance
$$\sigma^2 = \text{Var}(X) = \frac{n^2-1}{12}$$

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$$MGF \Rightarrow M(t) = \frac{e^t - e^{npt}}{n(1-e^t)}$$

* Binomial distribution

$$PMF \Rightarrow P(x) = \begin{cases} \binom{n}{x} p^x q^{n-x} & x=0,1,2,\dots,n \\ 0 & \text{elsewhere} \end{cases}$$

$$\text{Mean} \Rightarrow \mu = np$$

$$\text{Variance} \Rightarrow \sigma^2 = \text{Var}(X) = npq \quad \because q=1-p$$

$$MGF \Rightarrow M(t) = [(1-p) + pe^t]^n$$

* Negative Binomial distribution

$$PMF \Rightarrow \binom{x-1}{r-1} p^r q^{x-r}$$

$$\text{Mean} \Rightarrow \mu = \frac{r(1-p)}{p}$$

$$\text{Variance} \Rightarrow \sigma^2 = \frac{r(1-p)}{p^2}$$

$$MGF \Rightarrow \left(\frac{1-p}{1-pe^t} \right)^r$$

$$\text{Parameters} \Rightarrow [r \text{ and } p]$$

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* Geometric Distribution

$$\text{PMF} \Rightarrow P_X(x) = (1-p)^{x-1} p \quad x = 1, 2, 3, \dots$$

$$\text{Mean} \Rightarrow \mu = 1/p$$

$$\text{Variance} \Rightarrow \sigma^2 = \frac{1-p}{p^2}$$

$$\text{PMGF} \Rightarrow M(t) = \frac{pe^t}{1-(1-p)e^t} \quad \forall t < -\ln(1-p)$$

$$\text{Parameter} \Rightarrow p \text{ only}$$

* Multinomial distribution

$$\text{Mean} \Rightarrow E(X_i) = np_i$$

$$\text{Variance} \Rightarrow \text{Var}(X_i) = np_i(1-p_i)$$

* Hypergeometric distribution

$$\text{PMF} \quad P(X=x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$$

$$\text{Mean} \quad E(X) = \frac{nk}{N}$$

$$\text{Variance} \quad \text{Var}(X) = \frac{nk}{N} \left(\frac{N-K}{N} \right) \left(\frac{N-n}{N-1} \right)$$

$$\text{Parameters} \quad x, N, n, k$$

Poisson distribution

PMF $\Rightarrow P(X) = \frac{e^{-\lambda} \lambda^x}{x!}$

Mean $\mu = \lambda$

Variance $\sigma^2 = \lambda$

MGF $M(t) = e^{\lambda(e^t - 1)}$

Parameter λ only

Exponential distribution

Mean $\mu = 1/\lambda$

Variance $\text{Var}(X) = 1/\lambda^2$

Covariance $\text{Cov}(X) = 1$

Normal distribution

Mean $= \mu$

Variance $= \sigma^2$